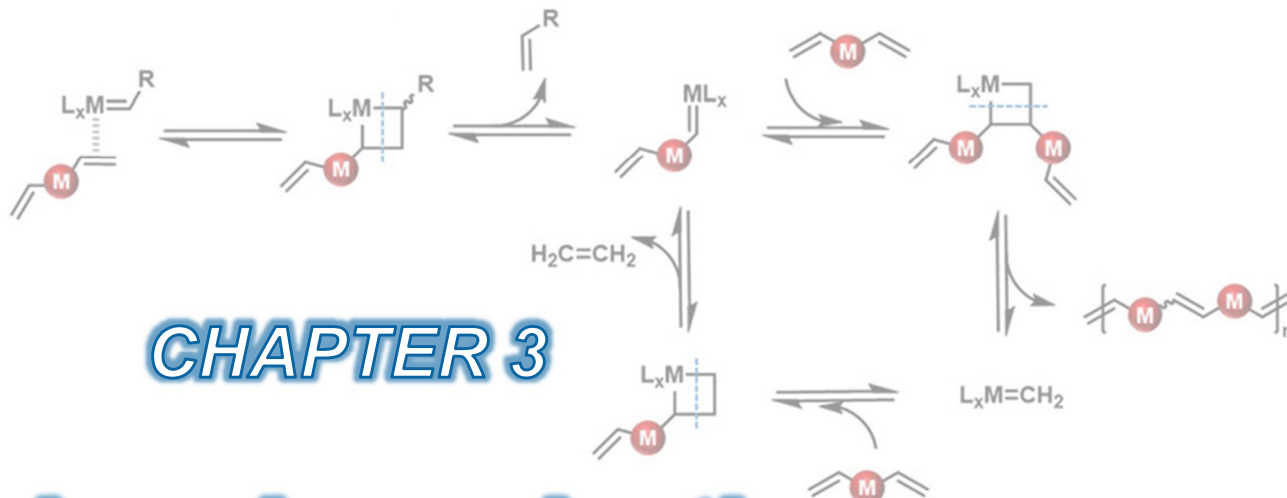
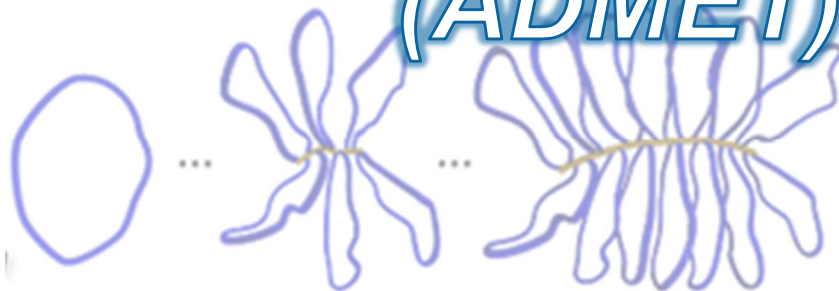


- Was versteht man unter Ziegler- bzw. Ziegler-Natta-Katalysatoren?
- Welche Reagenzien werden benötigt für die Synthese eines Ziegler-Katalysators?
- Warum funktioniert die radikalische Polymerisation von Propen nicht?
- Was heißt Taktizität? Wie viele Formen gibt es? Geben Sie Beispiele mit einem bestimmten Polymertyp.
- Welche zwei Methoden werden für die industrielle Synthese von Polyethylen angewendet? Welche PE-Typen werden so hergestellt?
- Geben Sie drei Beispiele von homogenen Katalysatoren, die eine bestimmte Taktizität bei PP liefern.
- Was sind COC's? Geben Sie eine Beispielstruktur an.



Cyclopolymerization and acyclic diene metathesis (ADMET) polymerization



J. Polym. Sci. Part A: Polym. Chem. **2000**, 38, 3451

Prog. Polym. Sci. **2017**, 69, 79

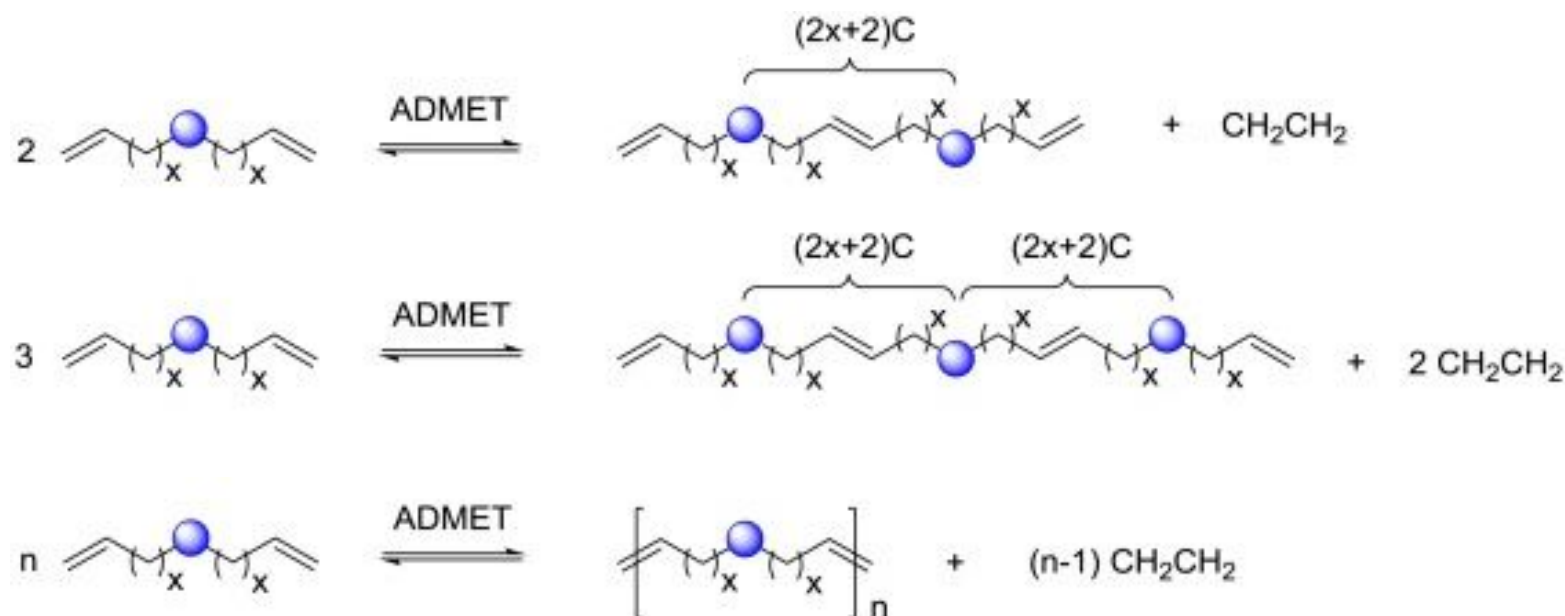
ADMET Polymerization

Definition

ADMET = Acyclic Diene Metathesis

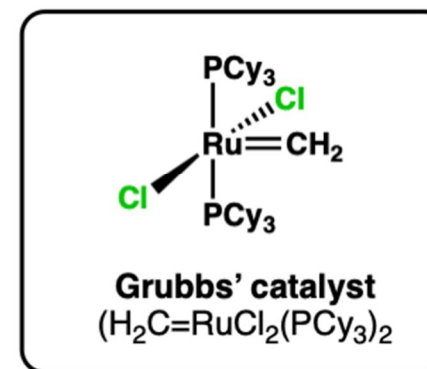
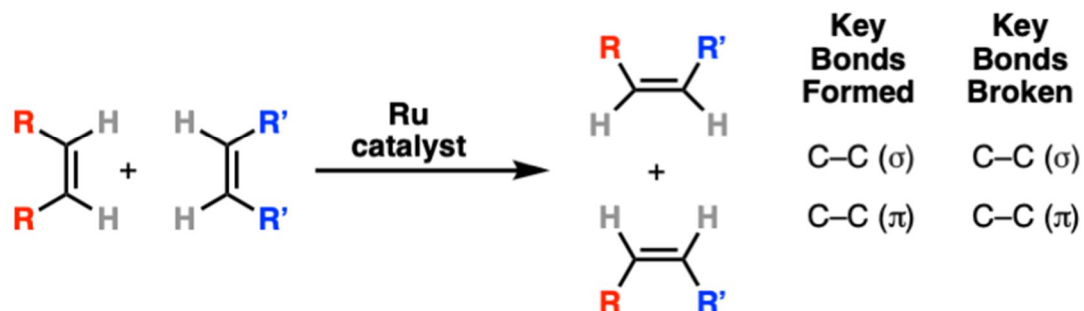
Growth type: **Step**

Monomer type: **α,ω -dienes**

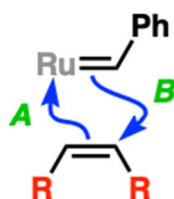


= exchange of alkene fragments

Olefin Metathesis

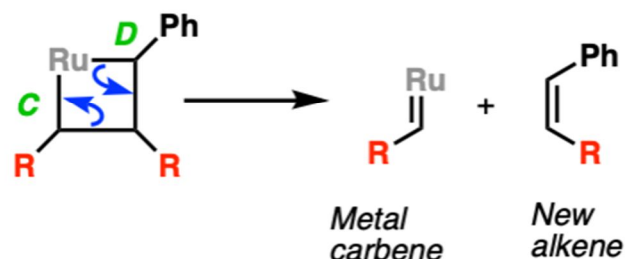


Step 1: Cycloaddition



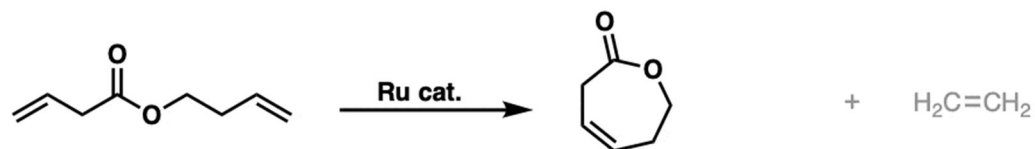
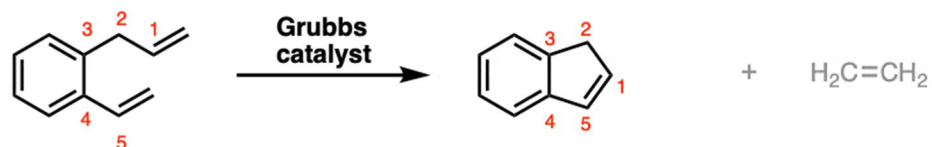
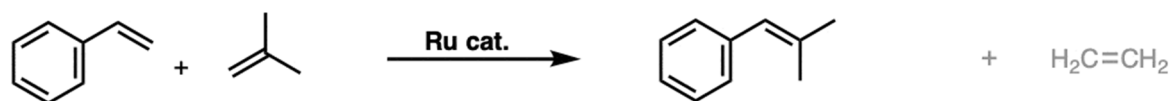
Ru is a simplified depiction of the Grubbs catalyst.

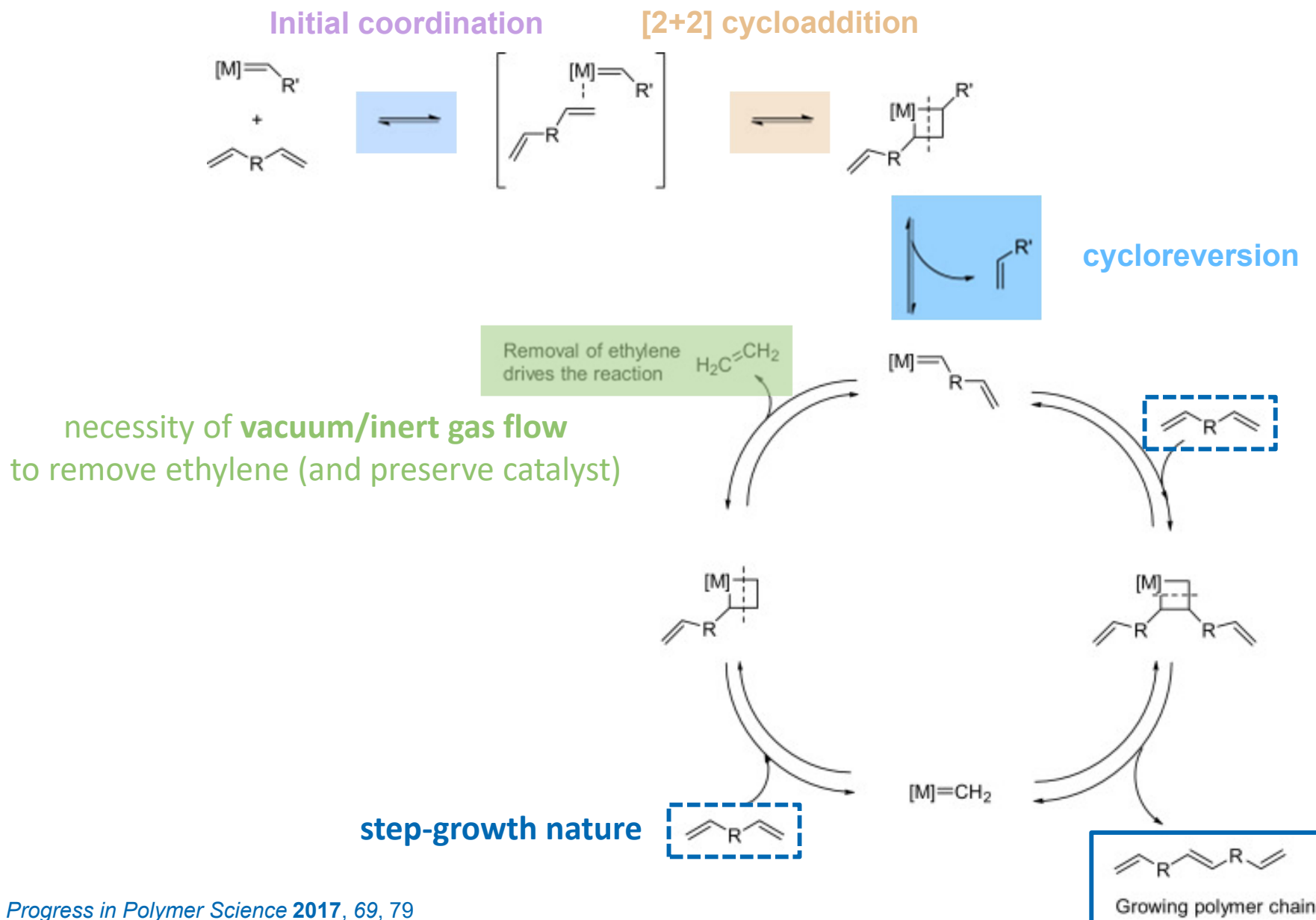
Step 2: Reverse cycloaddition



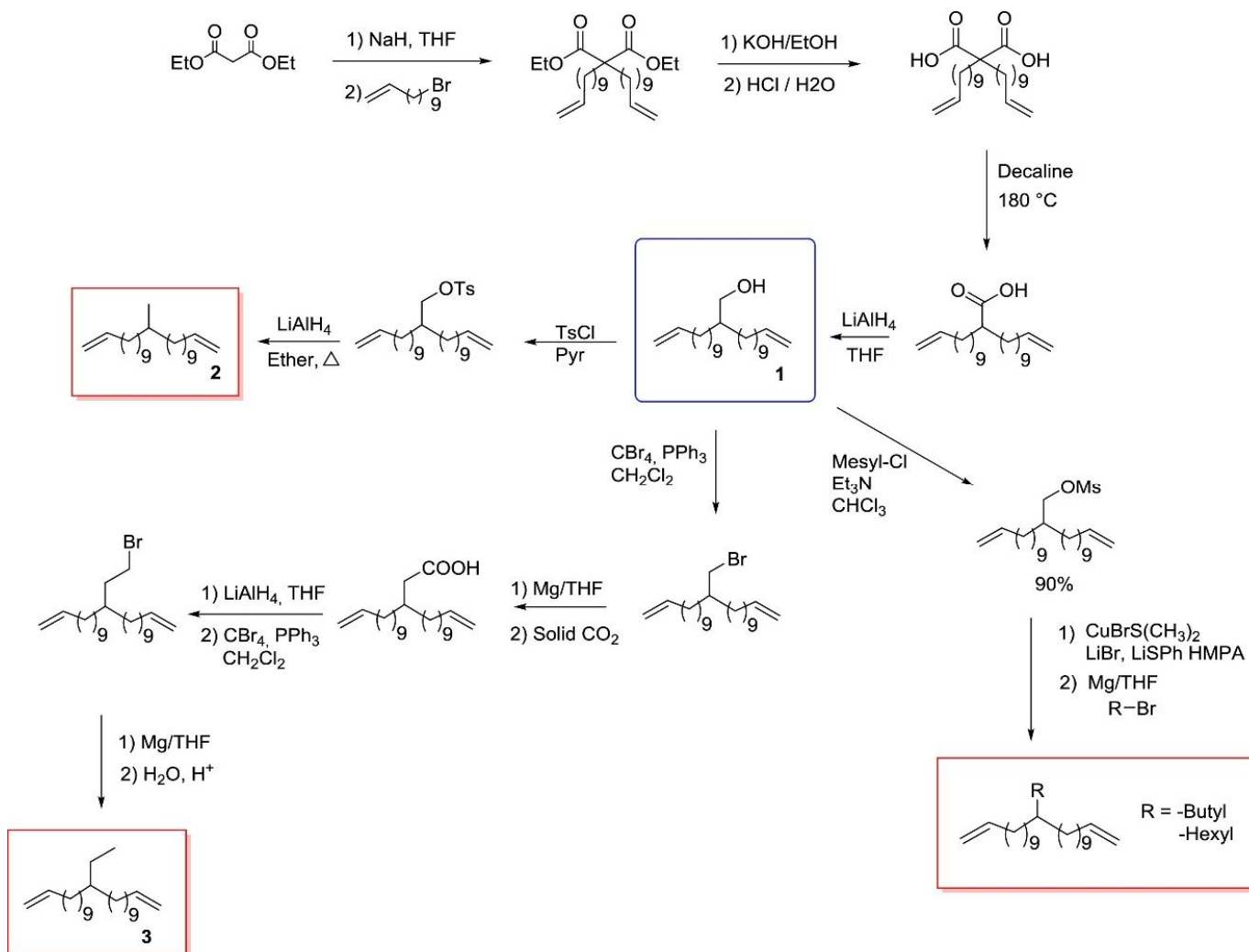
Bonds Formed	Bonds Broken
A C-Ru	A C-C (π)
B C-C	B Ru-C
C C-Ru	C C-C
D C-C (π)	D C-C (π) C-Ru

= exchange of alkene fragments

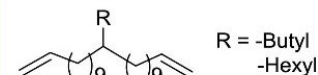
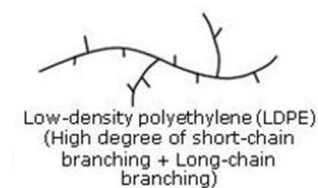
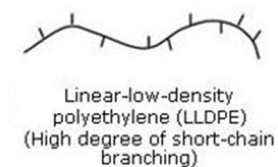
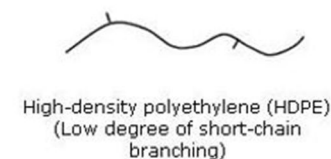


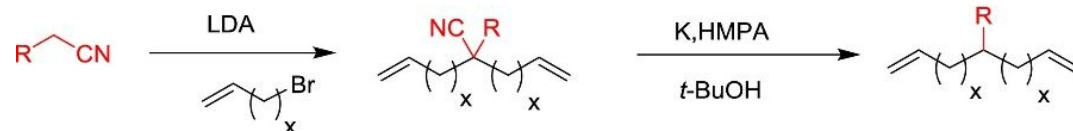


Practically, any α,ω -dienes as monomers can be polymerized.



For models of polyethylene





R = Methyl
Ethyl
Propyl
Butyl
Pentyl
Hexyl

Isopropyl
t-butyl
Cyclohexyl
Adamantyl

For models of polyethylene



High-density polyethylene (HDPE)
(Low degree of short-chain branching)



Linear-low-density polyethylene (LLDPE)
(High degree of short-chain branching)

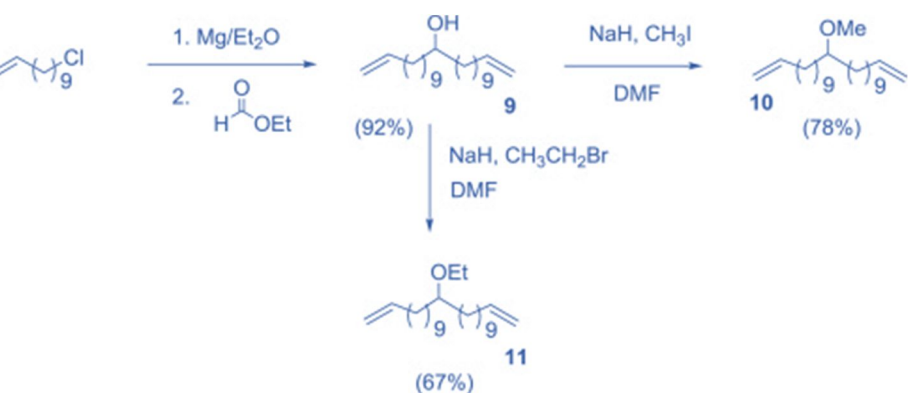
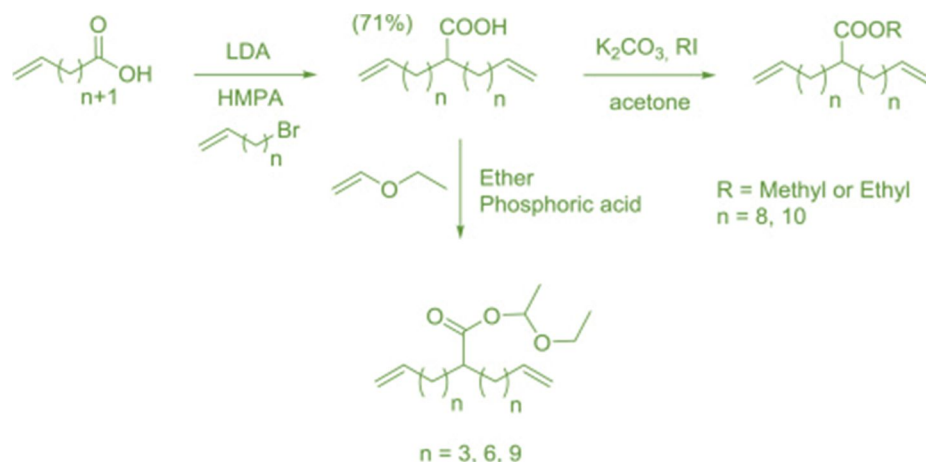
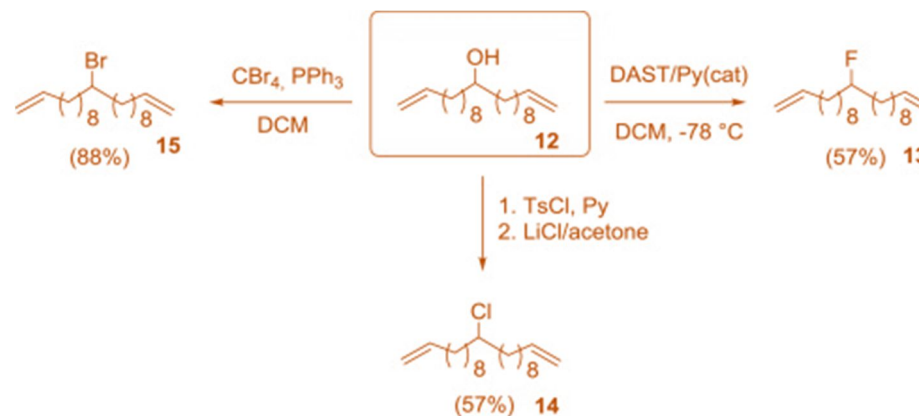


Low-density polyethylene (LDPE)
(High degree of short-chain branching + Long-chain branching)

Table 1. Functional group tolerance of early and late transition metal olefin metathesis catalysts
Activity increases up the table

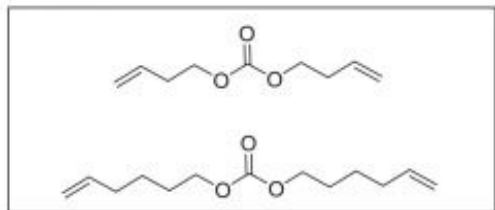
Titanium	Tungsten	Molybdenum	Ruthenium
Acids	Acids	Acids	Olefins
Alcohols, water	Alcohols, water	Alcohols, water	Acids, alcohols
Aldehydes	Aldehydes	Aldehydes	Water
Ketones	Ketones	Olefins	Aldehydes
Esters, amides	Olefins	Ketones	Ketones
Olefins	Esters, amides	Esters, amides	Esters, amides

Aust. J. Chem. **2005**, *58*, 3

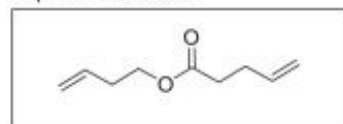


"In the first attempts to polymerize ester- and ether-containing monomers by ADMET it was observed that at least two methylene spacers from the carbonyl side of the ester were required for successful polymerization, presumably because the electrons on the oxygen would coordinate to the metal center during metathesis and inhibit the catalyst."

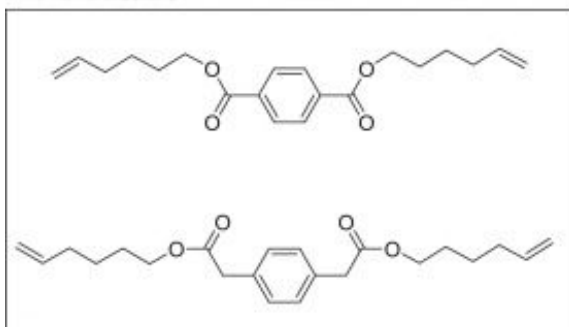
Carbonates



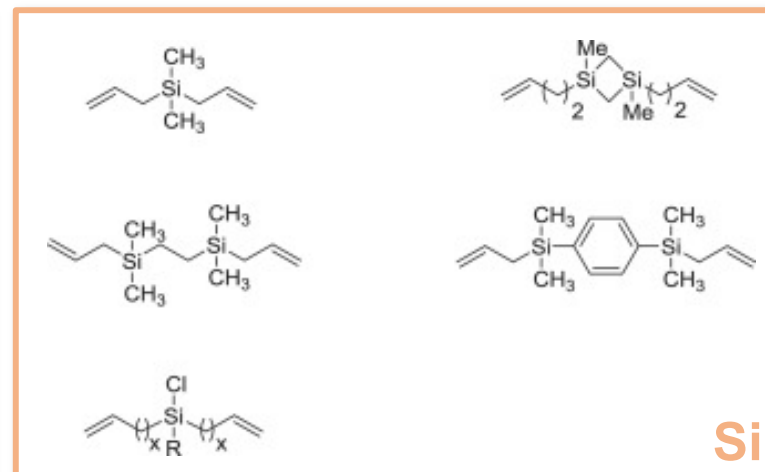
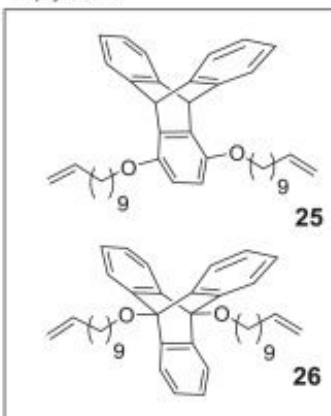
Aliphatic monoester



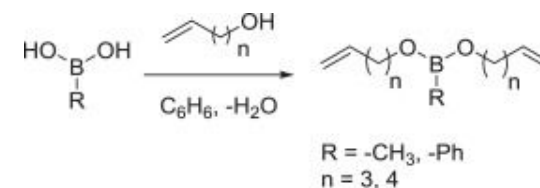
Aromatic diesters



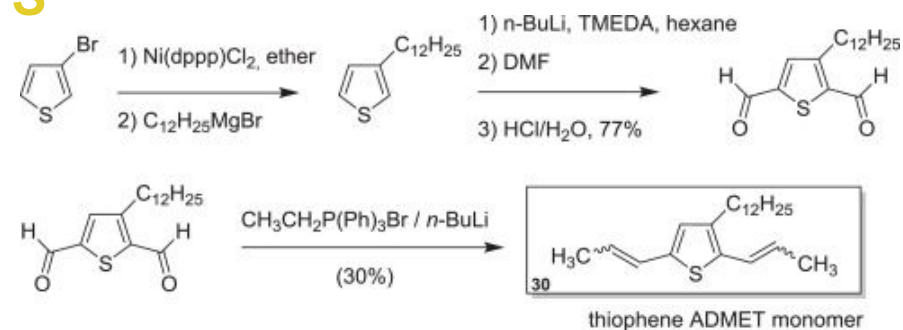
Triptycenes



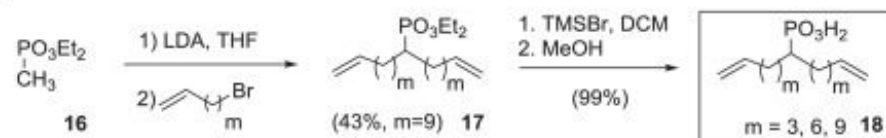
B



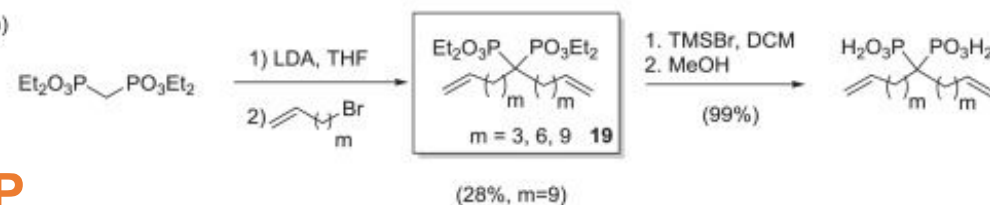
S



a)



b)

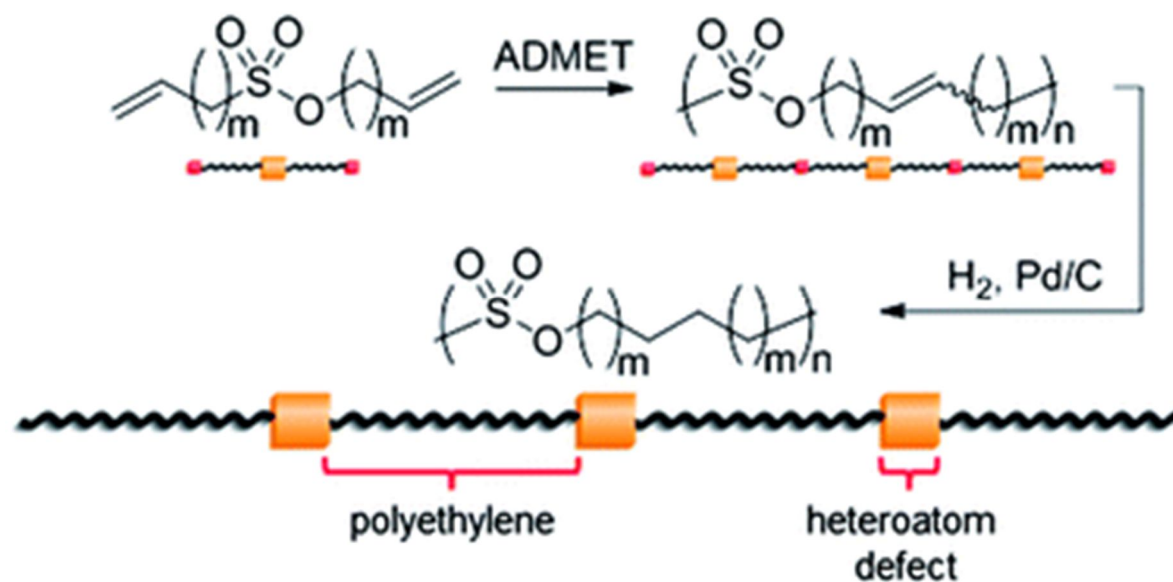


P

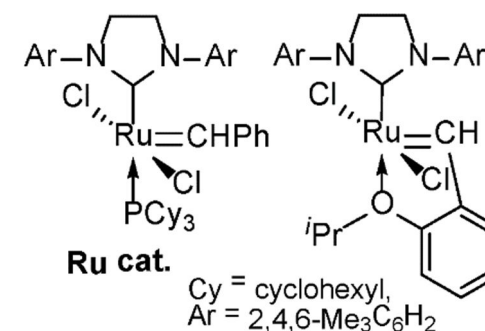
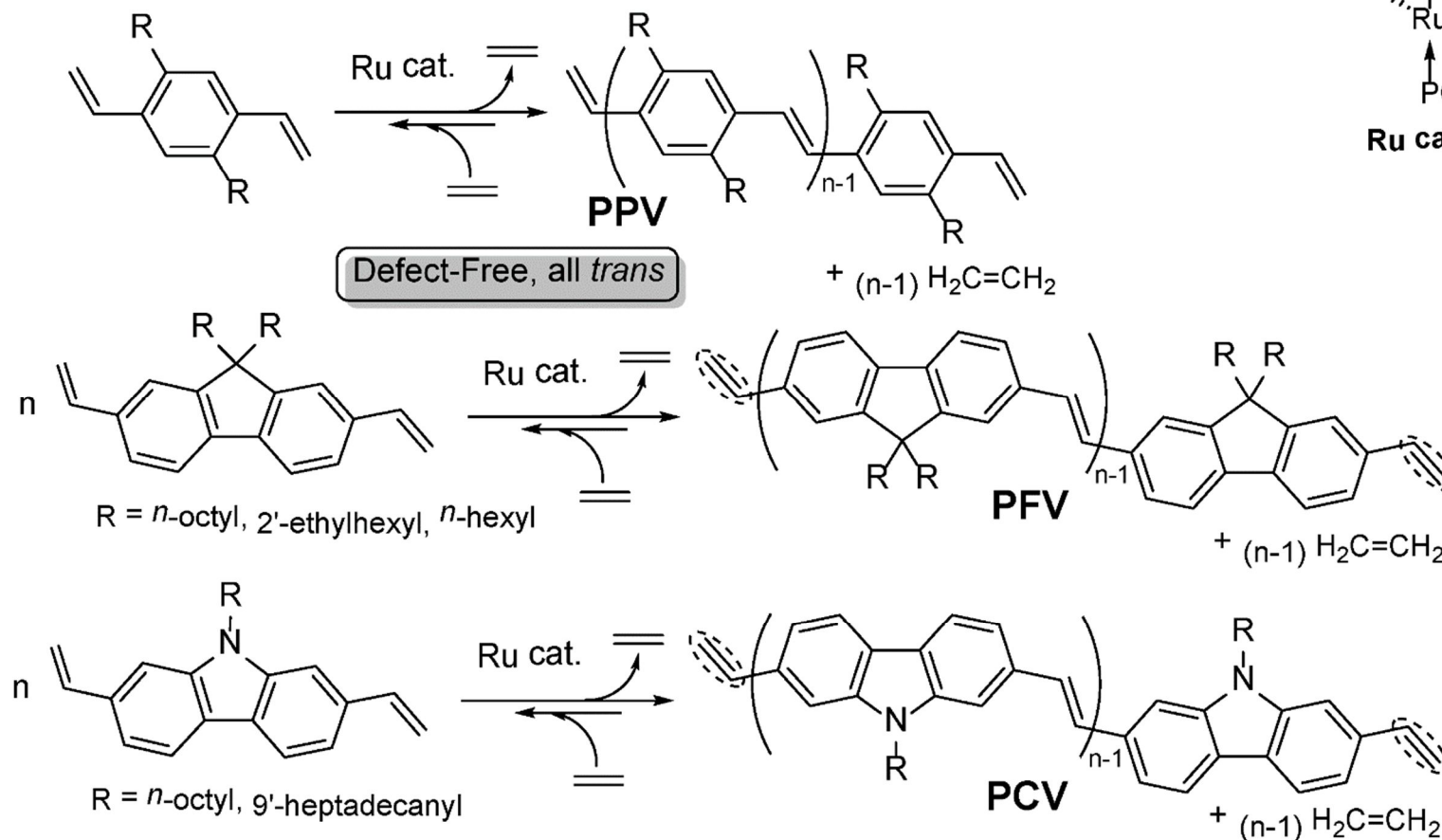
ADMET Polymerization

Applications and Advantages

Polyethylenes with “defects”



Acyclic Diene Metathesis (ADMET) Polymerization Route

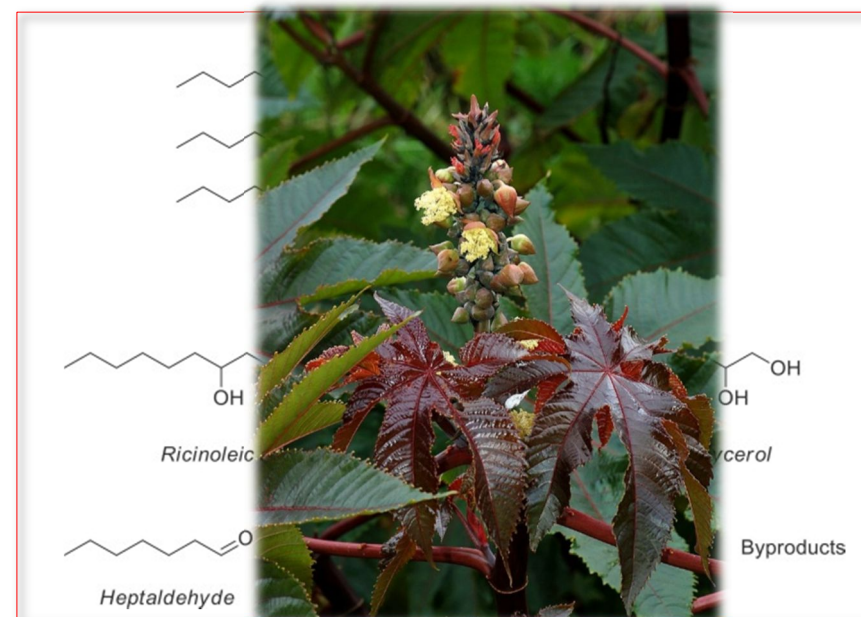
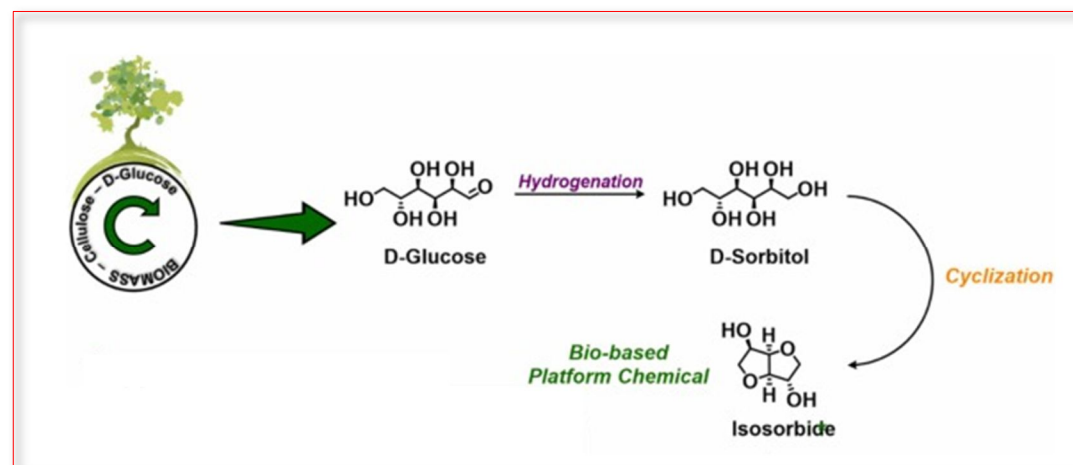
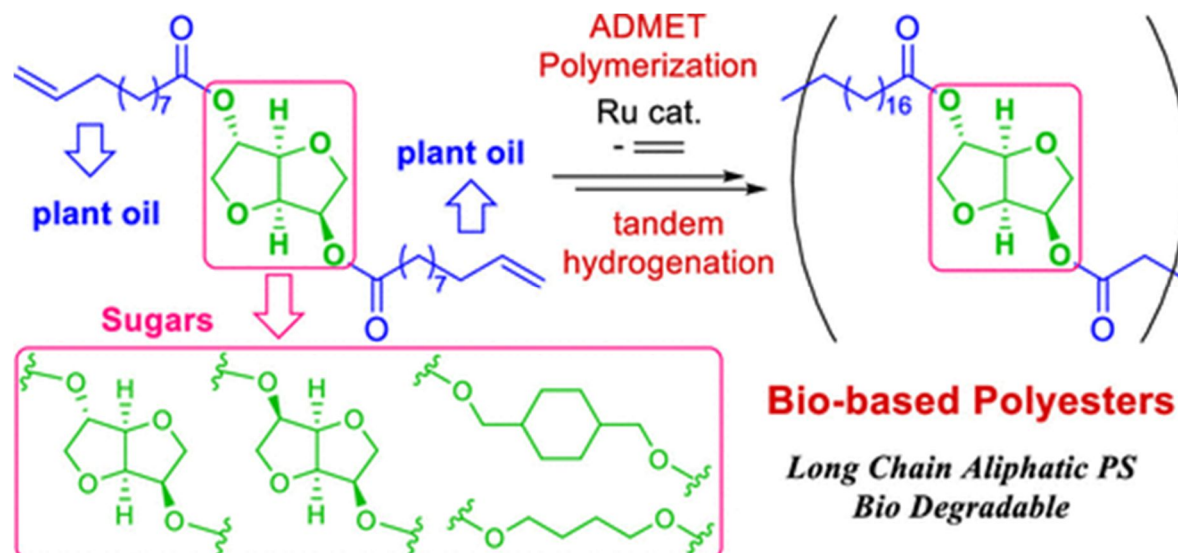


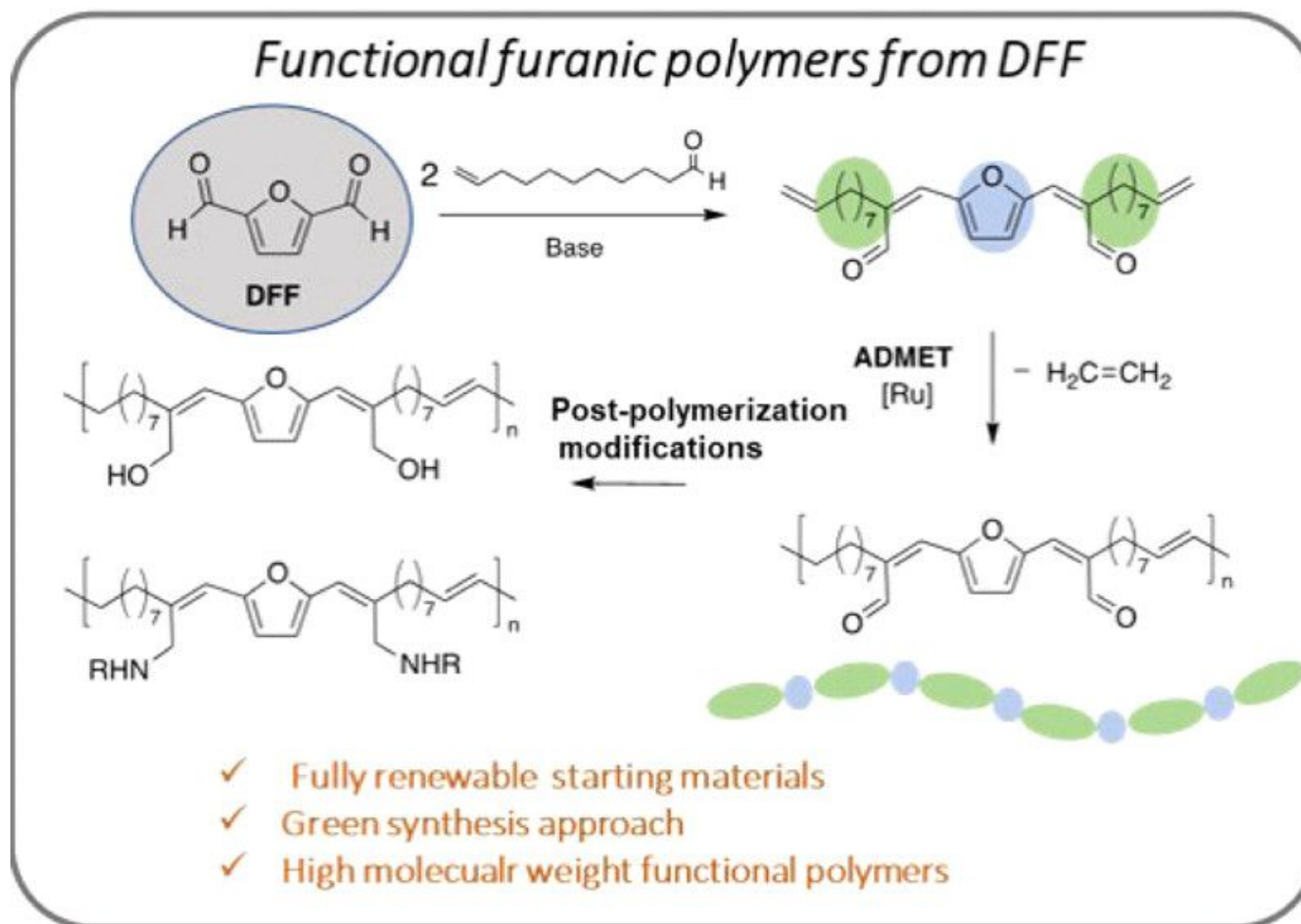
ADMET Polymerization

Applications and Advantages

Bio-based polymers

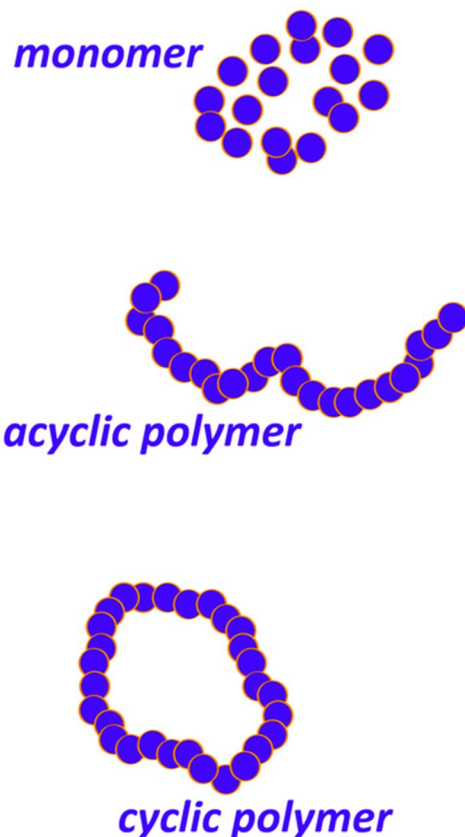
ACS Omega 2020, 5, 18301





- Synthesis of **well-defined, linear polymers** with **controllable functionality**.
- Use in **biodegradable polymers, liquid crystalline polymers, semiconducting materials**, etc.
- Advantages: No need for ring strain, functional group tolerance, structural precision.

- Limited to **diene monomers** (with sufficient spacing!)
- Step-growth polymerization → requires **pure monomers**
- Equilibrium → need to **remove ethylene effectively**
- **Interchange** reactions (can be an advantage too!)
- **Cyclization** reactions



Cyclic oligomers as by-products of step polymer(ization)s

Oligomers in poly(ethylene terephthalate) - Diffusion coefficients and migration

Maria Hoppe^a, Roberta Fornari^b, Pim de Voogt^c, Roland Franz^a

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Introduction

Polyethylene terephthalate (PET) is one of the most important food contact polymers. It is used for many food and beverage packaging applications due to its excellent properties like light-weight, flexibility, resistance to high temperature and low carbon dioxide permeability. Low molecular weight (LMW) oligomers can be formed during the synthesis process of PET as side products of incomplete polymerization^[1]. Other ways of oligomer formation can be external induced degradation processes through sunlight and food and beverage contact. Since these LMW oligomers are potential migrants into food and beverages^[2], it is of interest to know their diffusion characteristics in the polymer and hence their actual occurring migration levels.

Method

Experiments were carried out such that PET beverage bottle material was subjected to conditions simulating accelerated migration: 50% ethanol in water (v/v) for 15 h at 80°C using an 8üchi speed extractor (100 bar). The migration solutions were analysed every hour with LC-MS. The total oligomer content in the PET material was determined by extraction with dichloromethane.

Analyses were conducted using an Acquity UPLC system coupled to a Synapt G2Si (Waters).
Column: Acquity CSHTM Fluoro-Phenyl 1.7 µm 2,1x75 mm (Waters)
LC conditions: 0.5 ml/min, solvent A: methanol, solvent B: water, from 65% A to 100% A in 2.5 minutes, keep at 100% for 1 min and then to 65% in 0.1 min and keep at 65% for 1 min

Ionization: ESI positive
Mass range: 50 – 1200 m/z
External standard: cyclic PET trimer (LOQ = 0.1 ng/ml)

Results and discussion

Twelve different PET oligomers were identified and quantified in the migration solutions. Nine of them were cyclic, three of them were linear and assigned to the reaction products of cyclic oligomers with ethanol. Cyclic oligomers of a first series consisted of terephthalic acid (TPA) and ethylene glycol (EG), cyclic oligomers of a second series consisted of TPA, EG and one diethylene glycol (DEG) unit replacing an EG, in cyclic oligomers of a third series two EG units were replaced by two DEG units. Cyclic oligomers represent the biggest oligomer group (85%) besides the linear ones. The cyclic trimer of the first series and cyclic dimer of the second series were the most abundant oligomers. Five oligomers showed a Fickian migration behavior which is depicted in Figure 2 for the first series trimer and the second series dimer. Under the applied test conditions the measured values would correspond to a migration up to 0.07 mg/kg food for the first series trimer and 0.03 mg/kg food for the second series dimer.

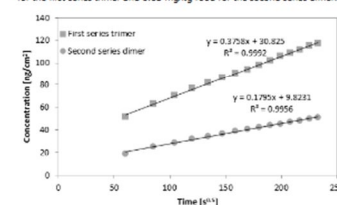


Figure 1: Migration kinetic of the first series PET trimer and second series PET dimer from a PET bottle into 50% ethanol at 80°C. Food simulant was replaced and analysed every hour for 15 cycles.

8th International Symposium on Food Packaging – Scientific Developments Supporting Safety and Innovation, 16.-18. November 2016, Barcelona



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Fraunhofer
IVV

	Bottle wall concentration [µg/g]	Experimental D [cm²/s]	Calculated D ^{Welle} [cm²/s]
First series cyclic oligomers			
Dimer	33.50	$5.85 \cdot 10^{-16}$	$3.71 \cdot 10^{-15}$
Trimer	2921.78	$5.62 \cdot 10^{-15}$	$1.57 \cdot 10^{-14}$
Second series cyclic oligomers			
Dimer	280.80	$1.39 \cdot 10^{-15}$	$8.47 \cdot 10^{-16}$
Trimer	156.52	$1.80 \cdot 10^{-15}$	$9.49 \cdot 10^{-17}$
Third series cyclic oligomers			
Dimer	65.00	$1.35 \cdot 10^{-15}$	$4.37 \cdot 10^{-16}$

Table 1: PET oligomer diffusion coefficients (D) determined experimentally (into 50% ethanol, at 80°C and 100 bar), calculated using the model of Welle^[3].

For the five oligomers showing Fickian behavior the experimental diffusion coefficients could be determined using the slope of the linear correlation between the concentration in the food simulant plotted against the square root of time and the total concentration of the oligomers in the polymer. The other oligomers identified in the material stopped showing migration after the first or second cycle. Additionally the diffusion coefficients of the oligomers were predicted using the equation published by Welle^[3] which was developed especially for diffusion in PET. The values are shown in Table 1.

Diffusion coefficient values calculated with the equation by Welle were by 99.3 to 99.7% lower than the experimental D values which can be attributed to the swelling ability of ethanol for PET which causes a higher migration rate. However, calculating the diffusion coefficients using the Fick equation which is a conservative approach gives by 99.0 to 99.6% higher diffusion coefficient values than the experimentally determined ones. Even taking into account the swelling effect of ethanol, the experimental diffusion coefficients fit better with the model of Welle. Therefore the Welle equation might be a simple and fast way to calculate the migration of PET oligomers conveniently at very low migration levels which otherwise would not be easily directly accessible by analytical means.

References

- [1] Benhoun, J. M.; Choi, K. Y. (1989). "Identification and Characterization of Reaction by-Products in the Polymerization of Polyethylene Terephthalate." *Journal of Macromolecular Science-Reviews in Macromolecular Chemistry and Physics C29(1)*: 55-81.
- [2] Begley, T. H.; Dennison, J. L.; Hollifield (1990). "Migration into food of polyethylene terephthalate (PET) cyclic oligomers from PET microwave susceptor packaging." *Food Additives & Contaminants* 7(6): 797-805.
- [3] Welle, F. (2013). "A new Method for the Prediction of Diffusion Coefficients in Poly(ethylene terephthalate)", *Journal of Applied Polymer Science* 129(4): 1845-1851.

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maria.hoppe@ivv.fraunhofer.de

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Twelve different PET oligomers were identified and quantified in the migration solutions. Nine of them were cyclic, three of them were linear and assigned to the reaction products of cyclic oligomers with ethanol.

Why?

Because we can
and also because their physical properties differ from linear counterparts.

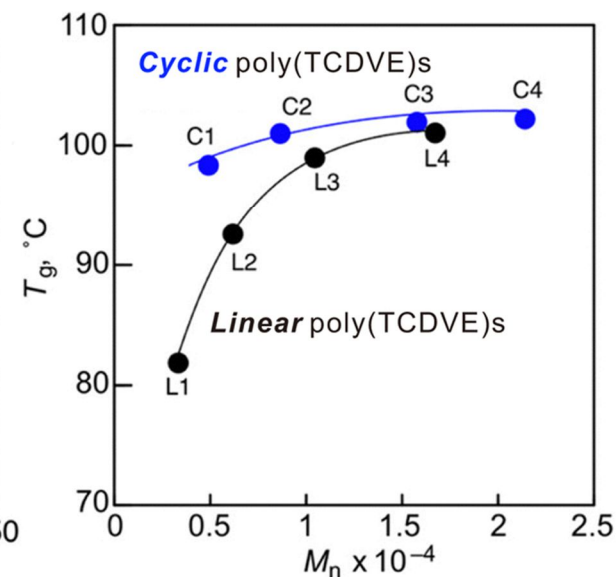
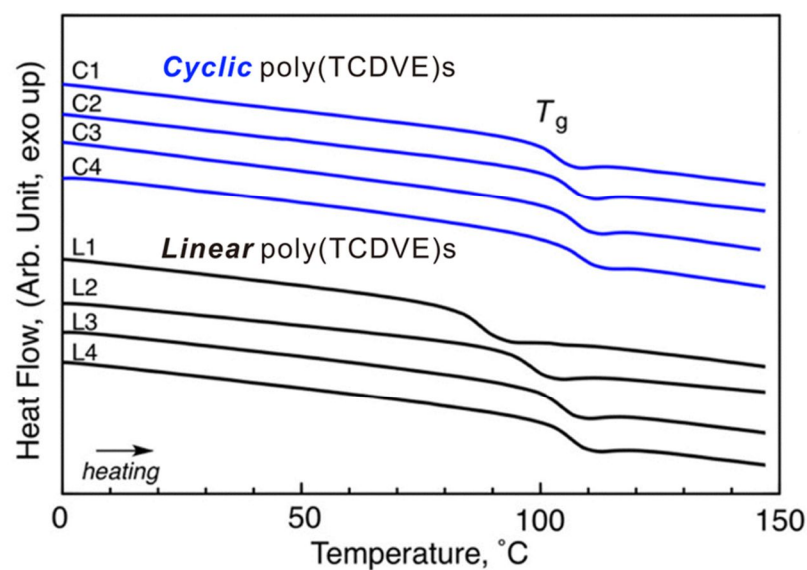
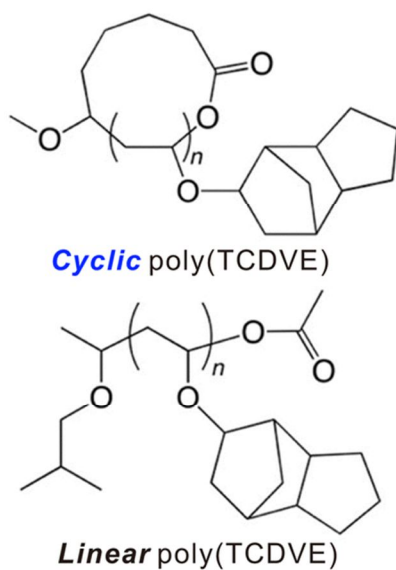
No chain ends!!! → a range of unexpected physical properties

Examples:

- reduced hydrodynamic size
- improved stability
- slower degradation profile
- clear differences in crystallization and glass transition behaviors as well as rheological properties
- distinct self-assembly behaviors both in solution and in thin films

→ Interesting for various application, e.g., drug delivery, surface modification, hierarchical assembly

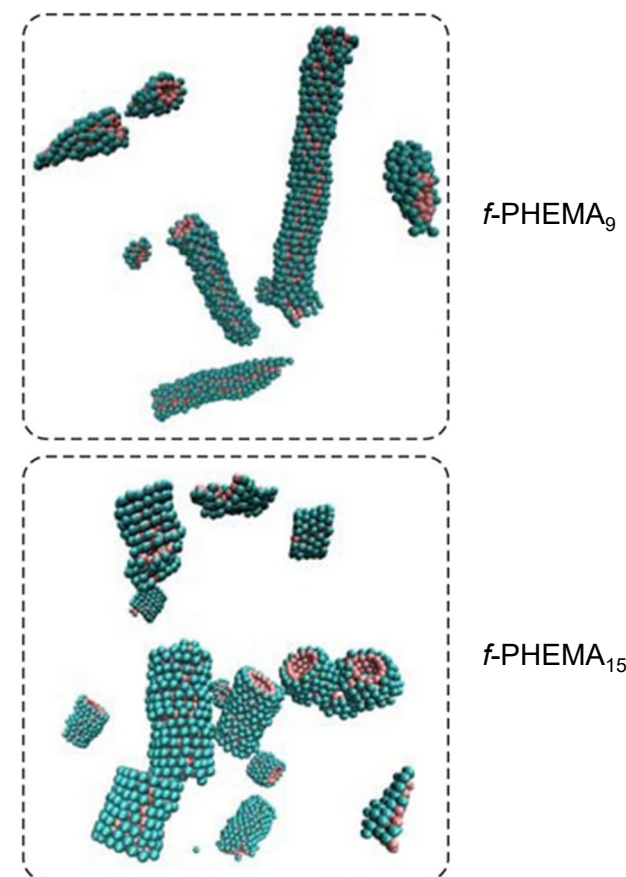
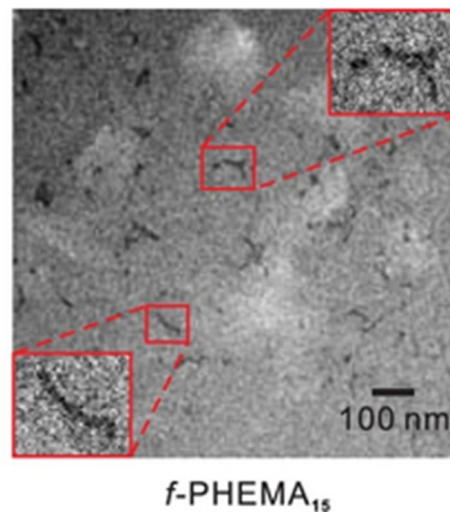
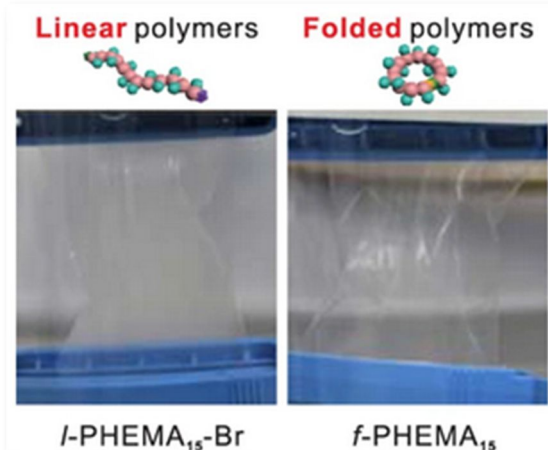
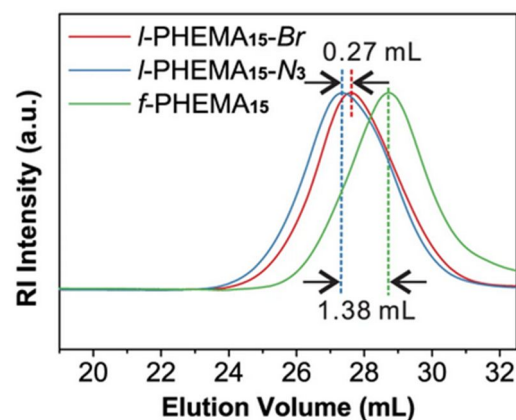
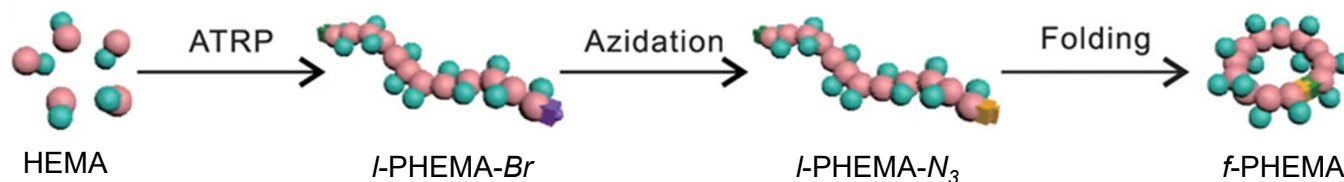
Physical properties different from linear counterparts



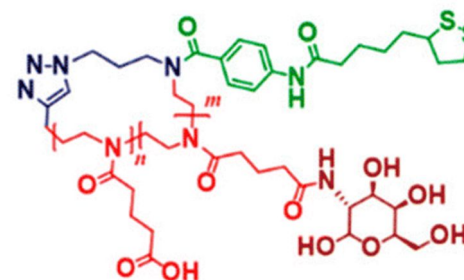
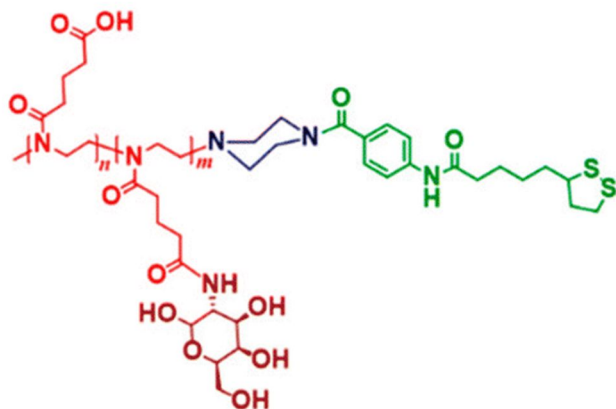
Cyclic Polymers

Why?

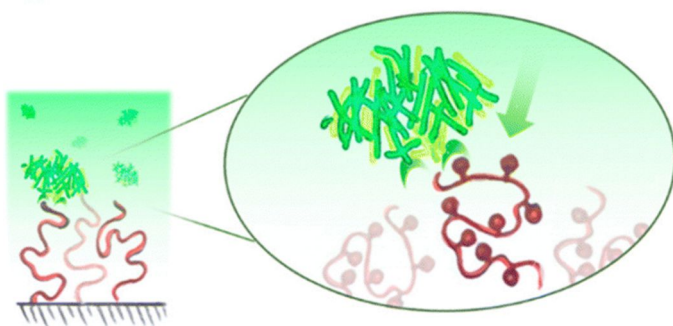
Physical properties different from linear counterparts



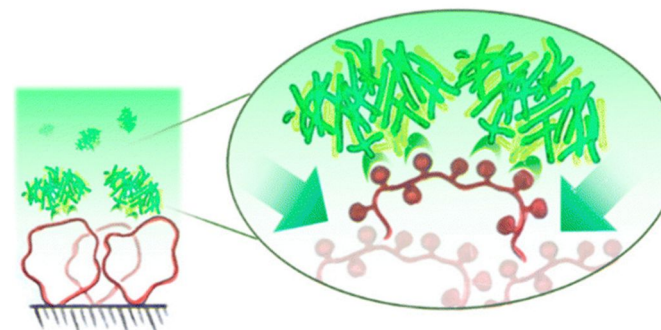
Nat. Commun. 2021, 12, 3959



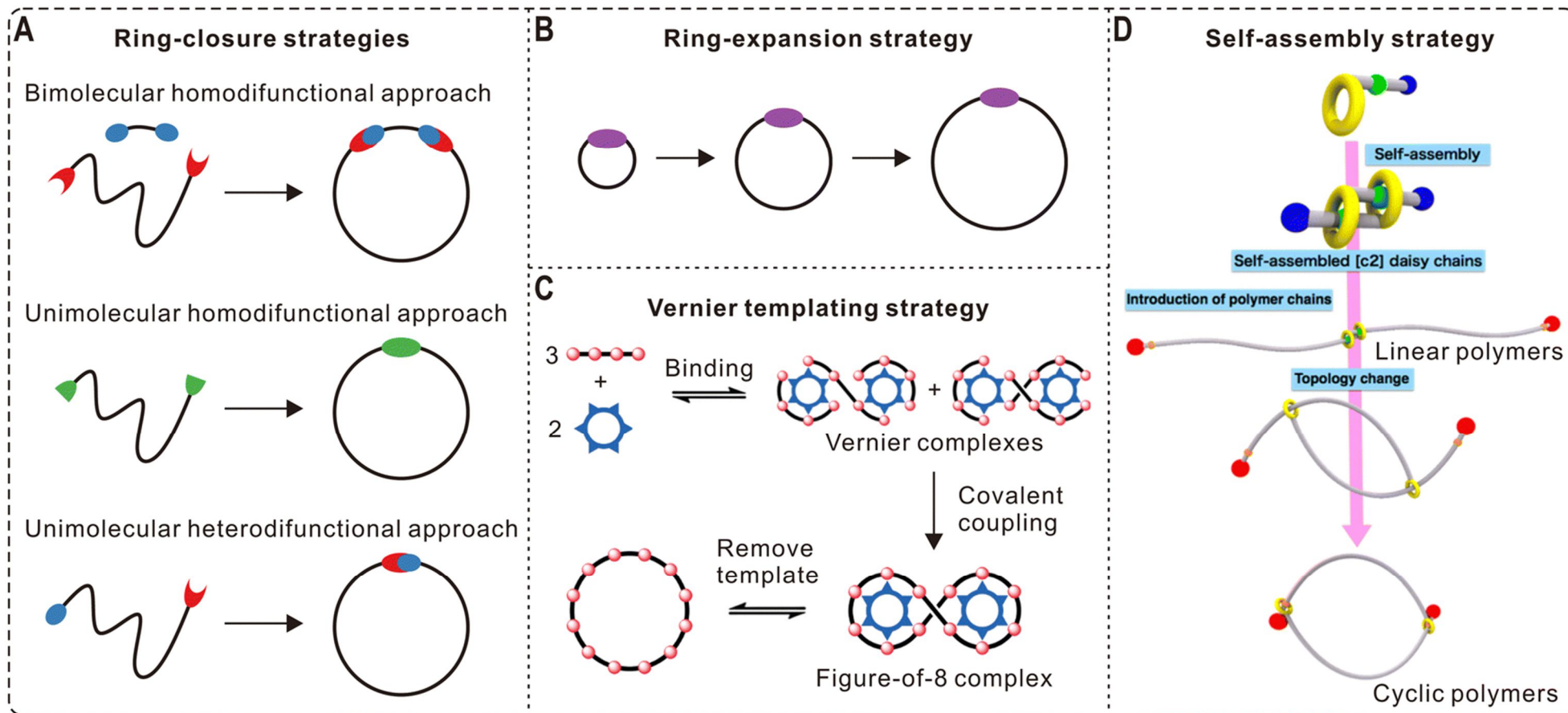
vs.



linear

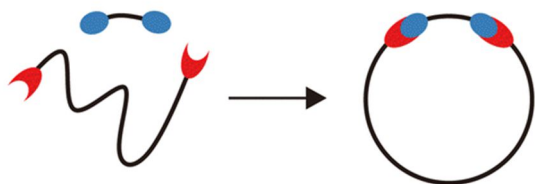


cyclic

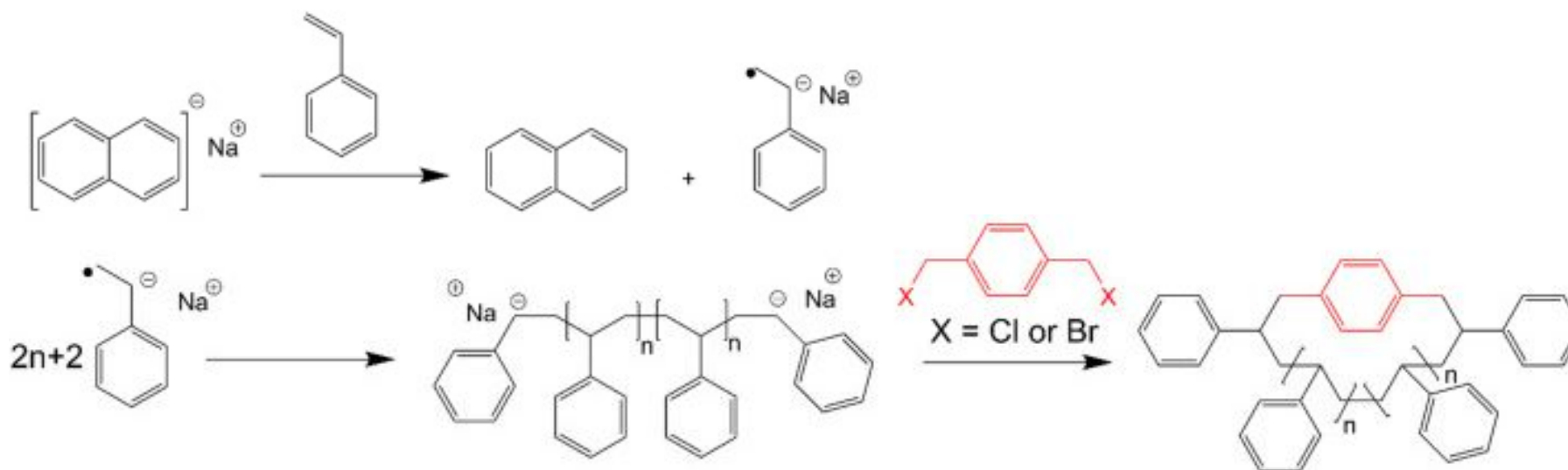


Ring-closing approach

Bimolecular homodifunctional approach

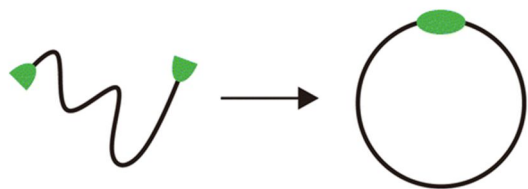


Example *bidirectional anionic polymerization, followed by in-situ double nucleophilic substitution*

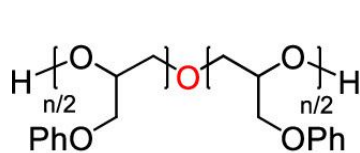


Ring-closing approach

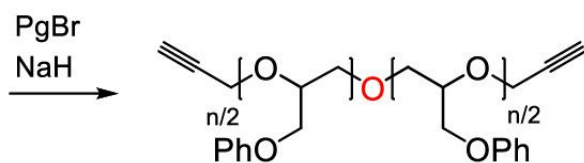
Unimolecular homodifunctional approach



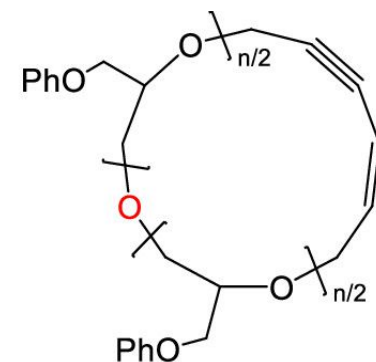
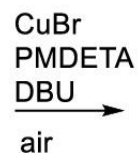
Example *bidirectional anionic polymerization, followed by end-group modification and Glaser coupling*



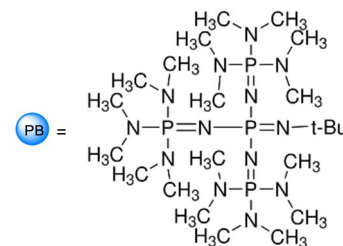
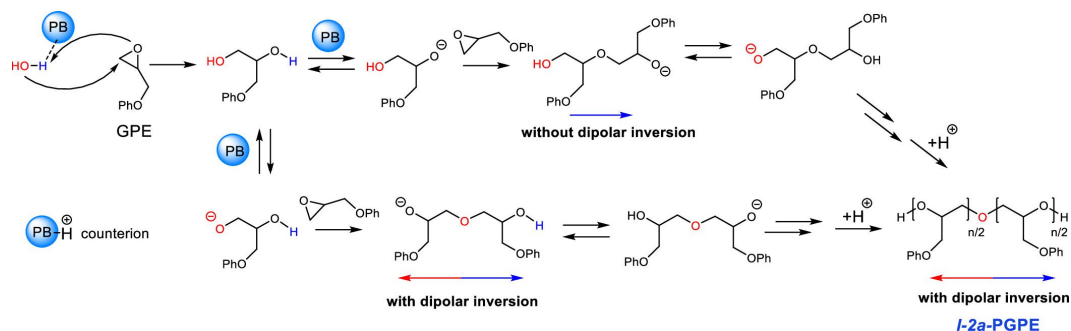
I-2a-PGPE



propargylated I-2a-PGPE

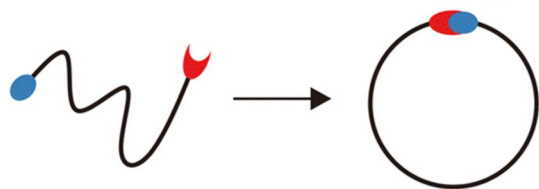


c-2a-PGPE

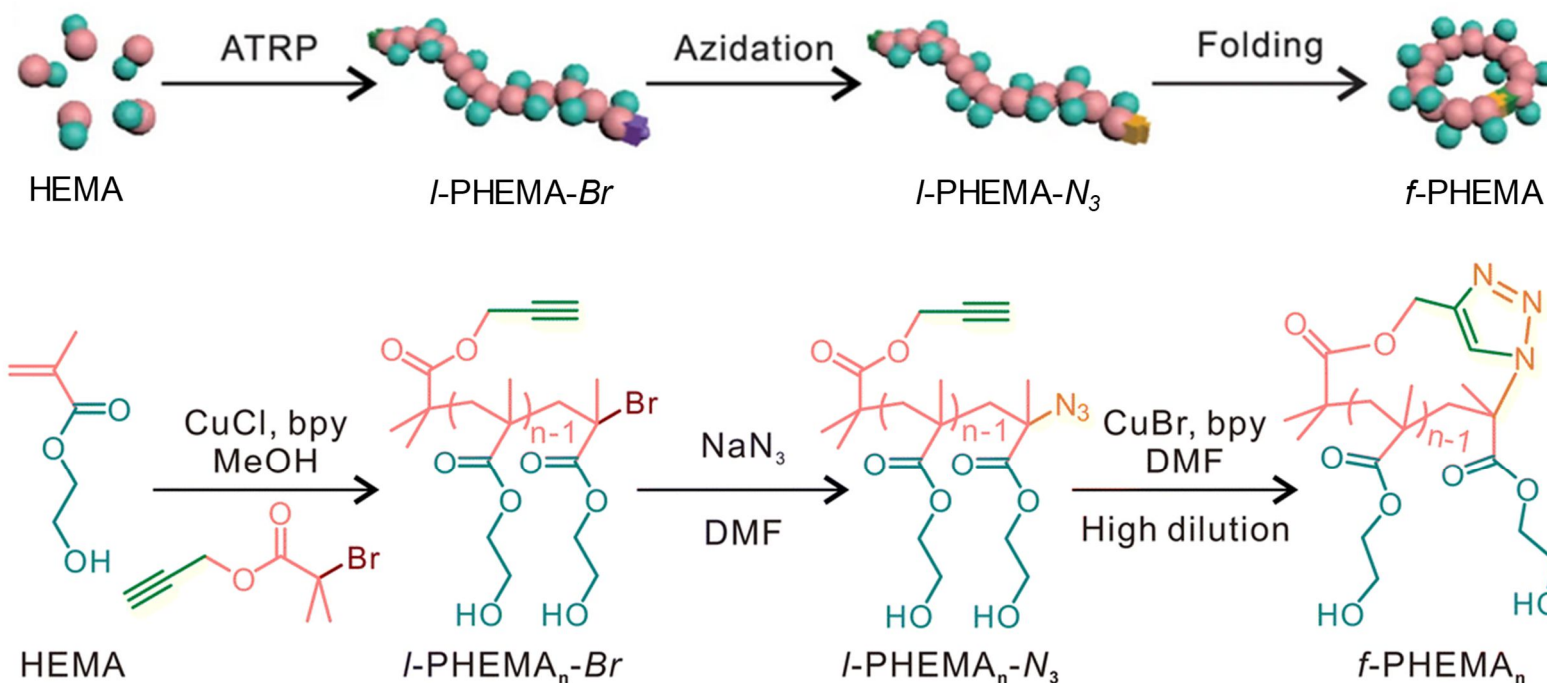


Ring-closing approach

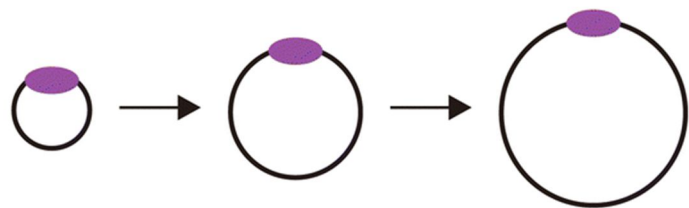
Unimolecular heterodifunctional approach



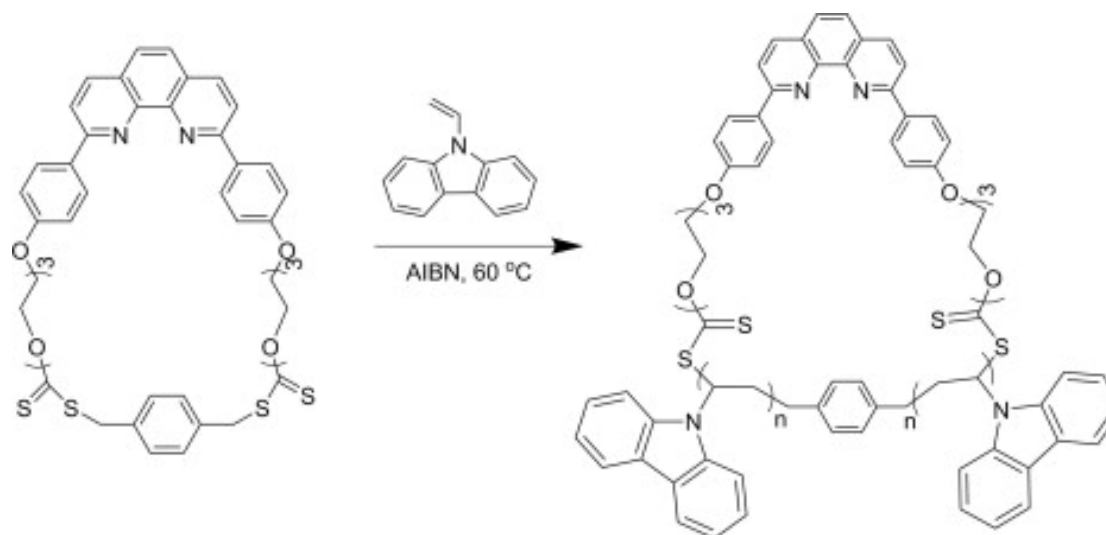
Example *ATRP from a functional initiator, followed by end-group modification and dipolar cycloaddition*



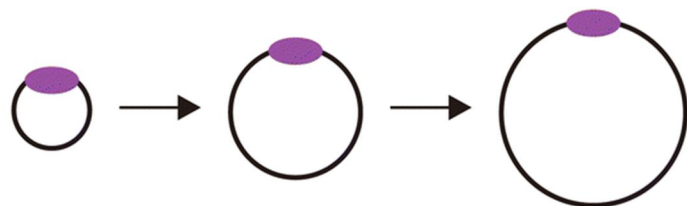
Ring-expansion approach



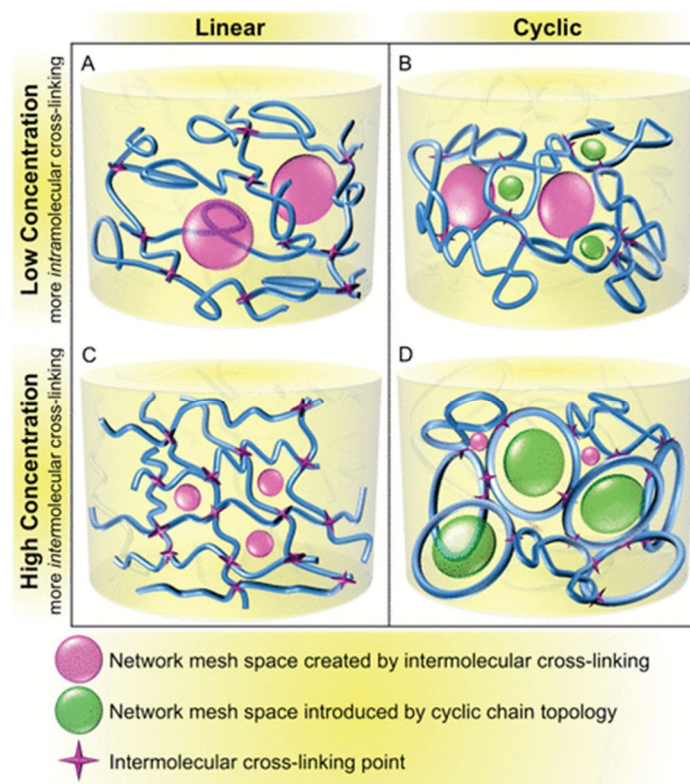
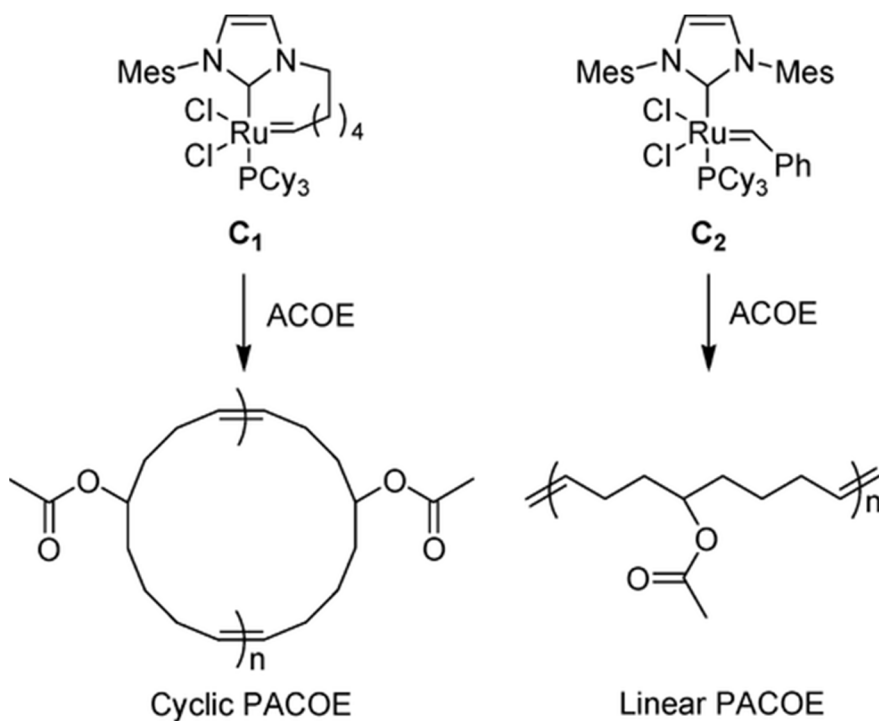
Example ring-expansion **RAFT** polymerization



Ring-expansion approach

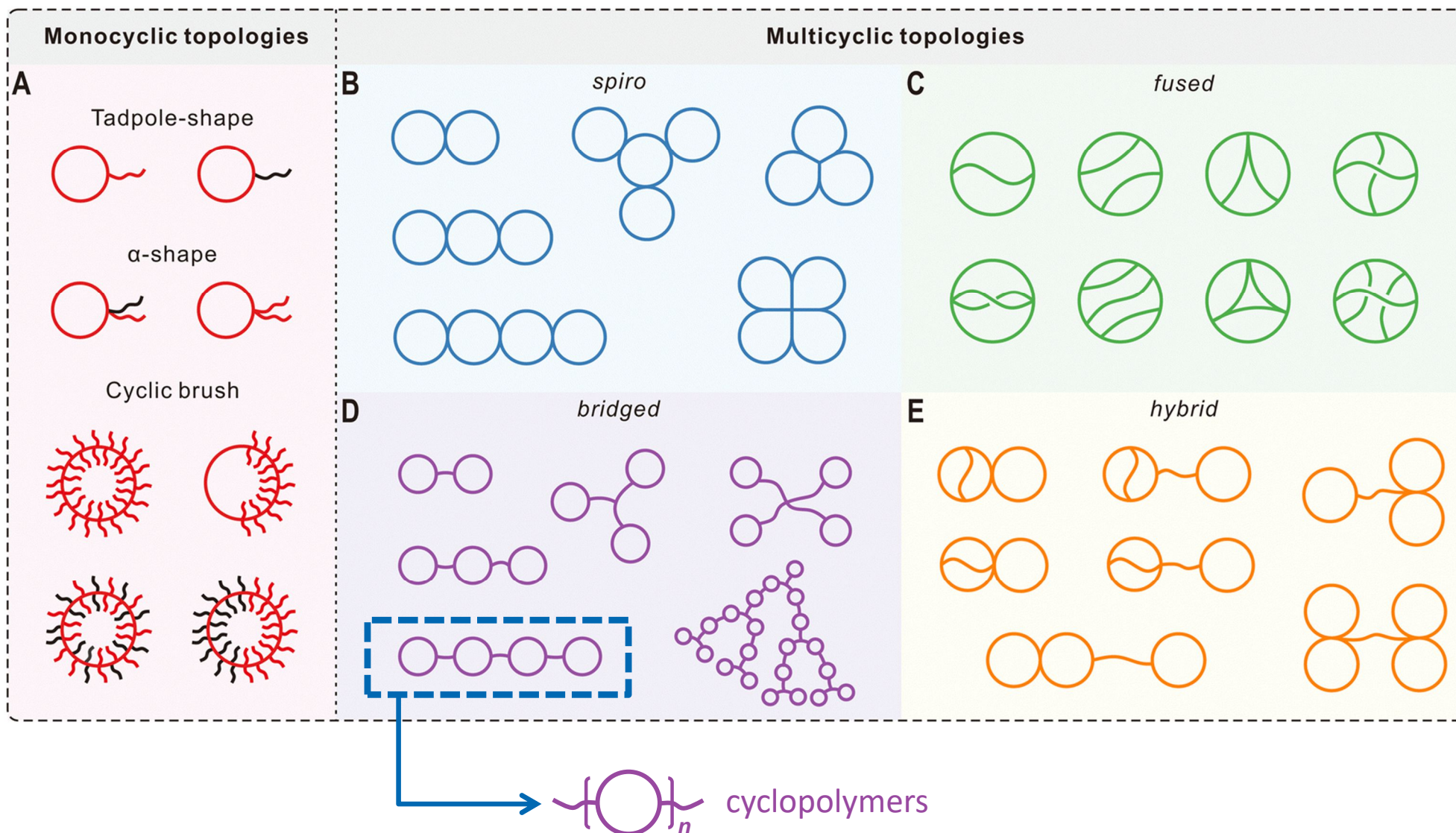


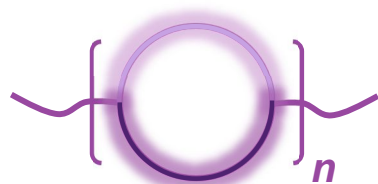
Example ring-expansion **metathesis polymerization**



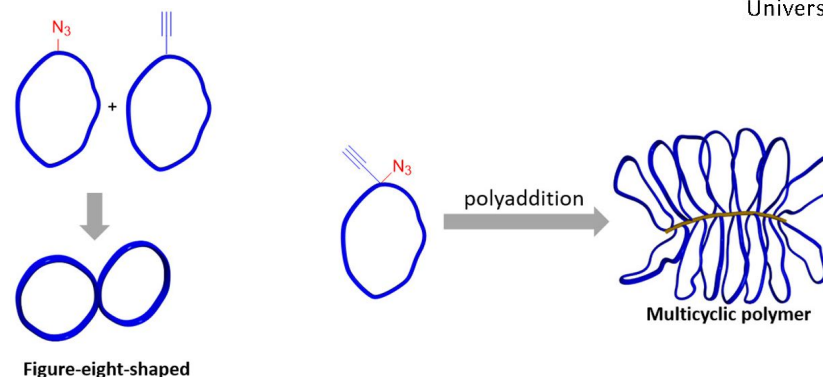
ACOE = 5-acetoxy-1-cyclooctene

J. Am. Chem. Soc. **2011**, *133*, 4140

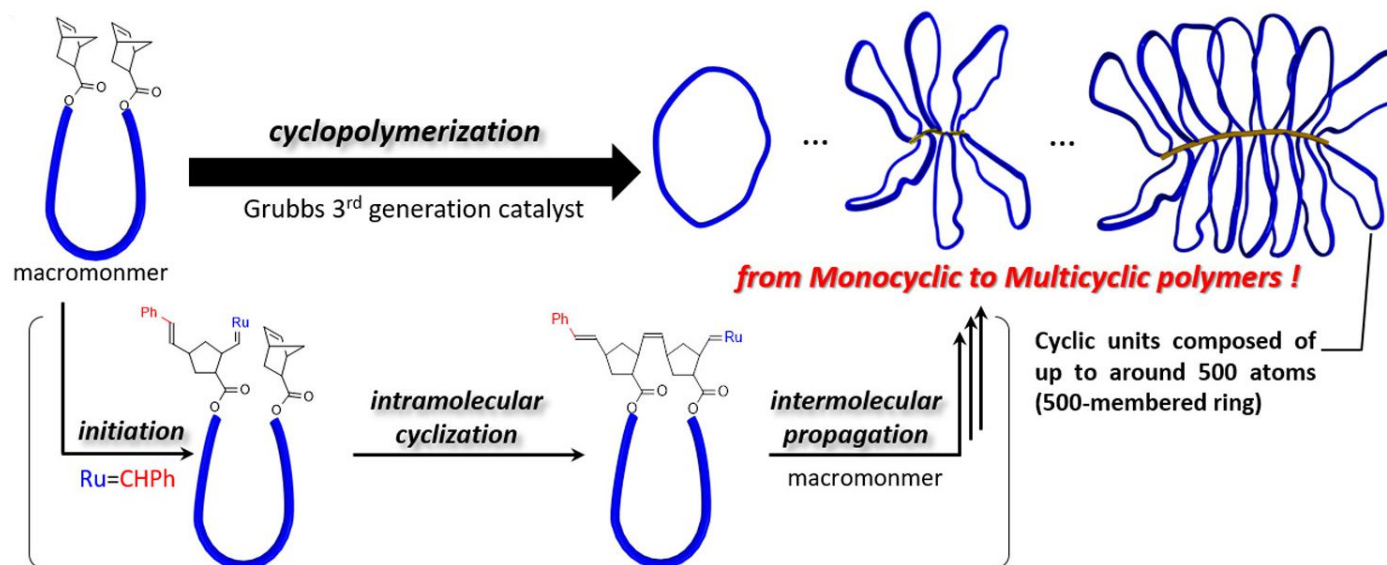




These **two parts** are usually not the same size.



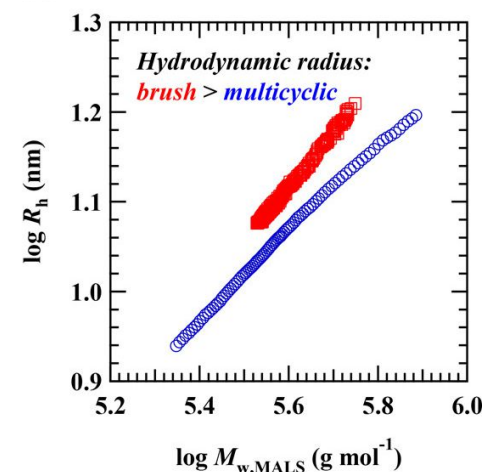
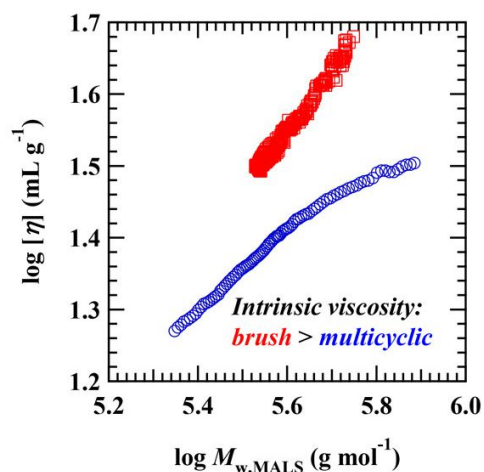
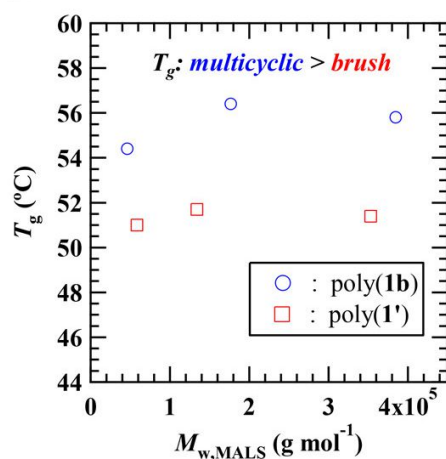
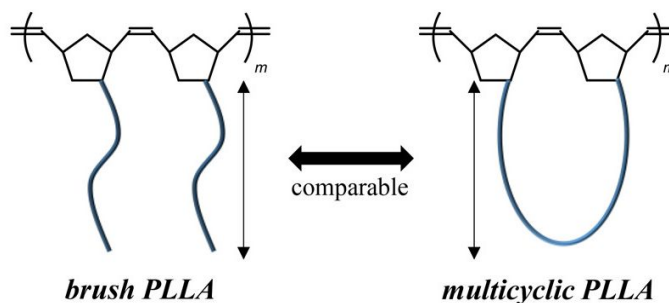
the organic chemist way of **coupling rings**



Above is a peculiar example: a cyclopolymer composed of **multiple cyclic polymers**. But cyclopolymers usually consist of “small” rings attached onto a main backbone.

Why?

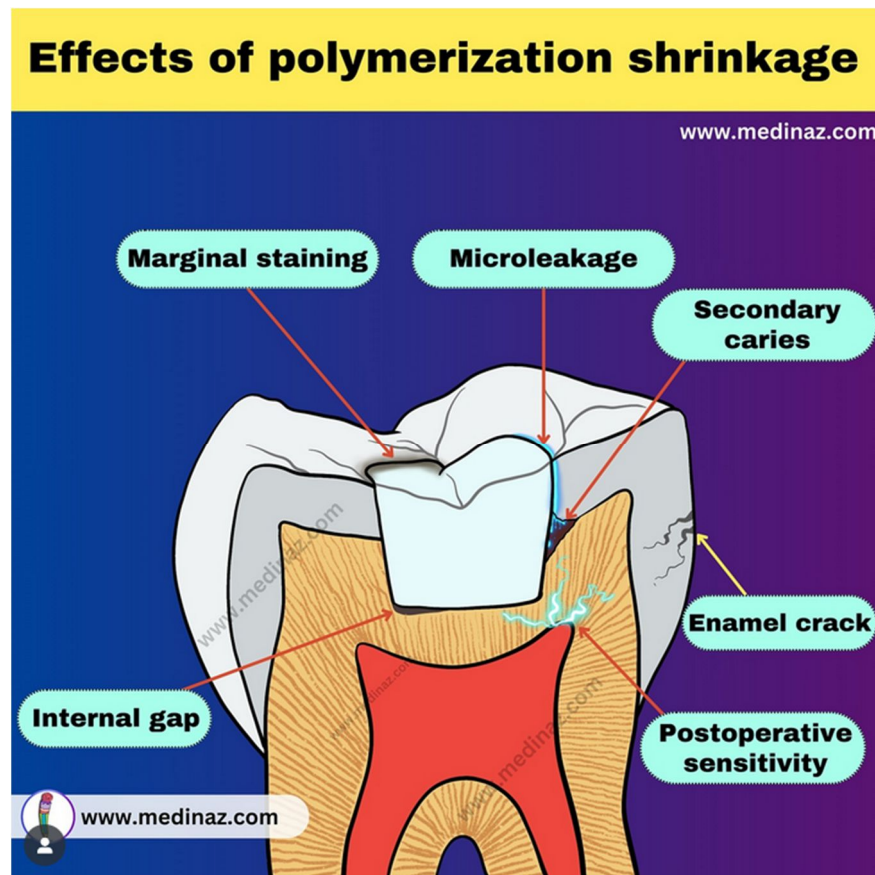
Because we can
and also because their physical properties differ from linear counterparts.



Why?

In addition, particularly for small rings, additional properties may appear:

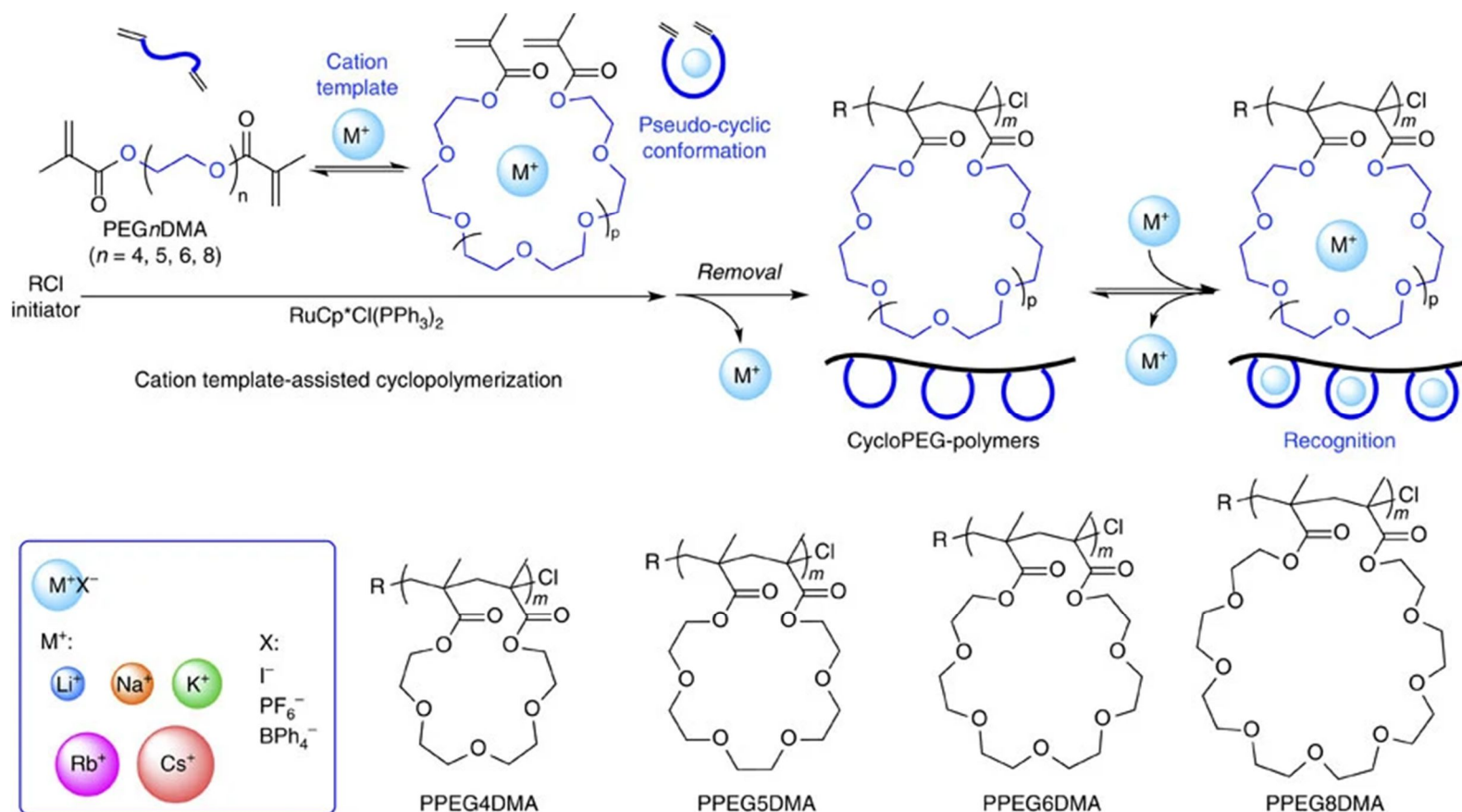
- reduced shrinkage (volumic contraction) during polymerization



Why?

In addition, particularly for small rings, additional properties may appear:

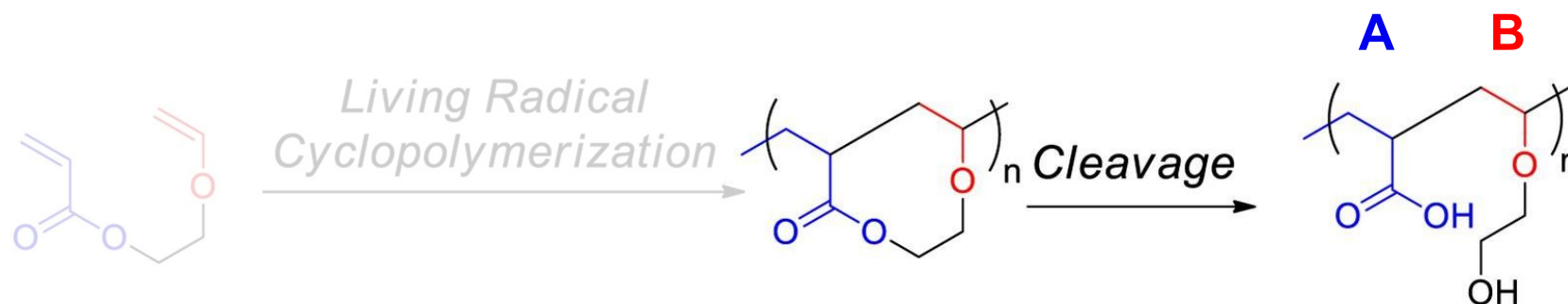
- reduced shrinkage (volumic contraction) during polymerization
- supramolecular complexation via macrocyclic effect



Why?

In addition, particularly for small rings, additional properties may appear:

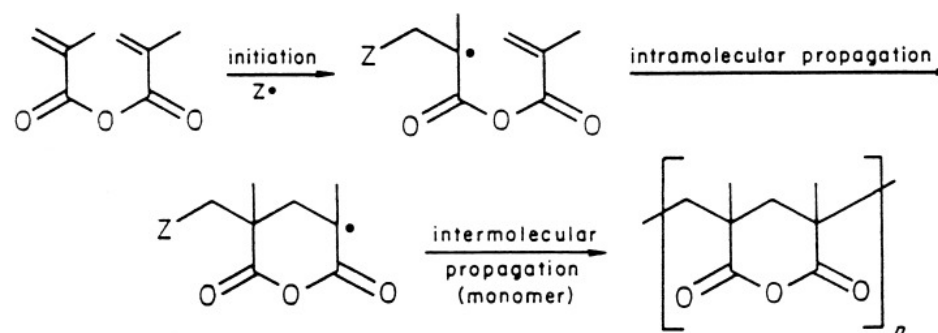
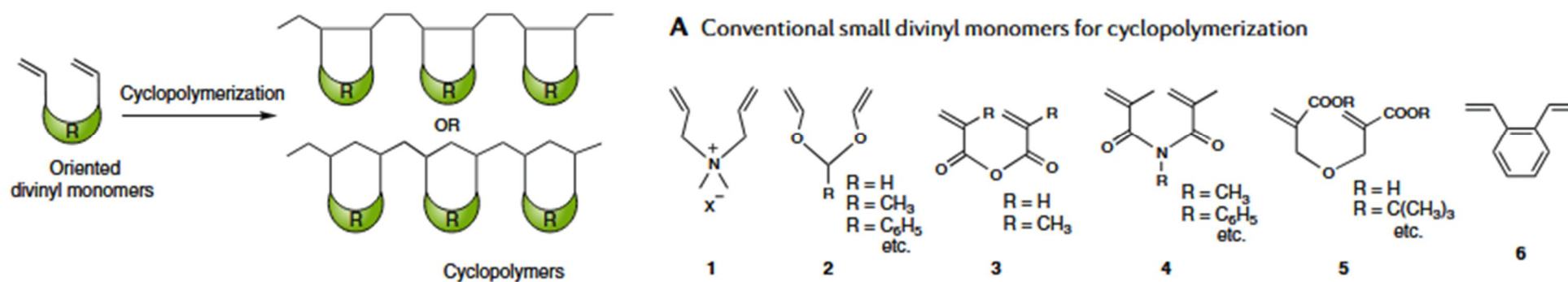
- reduced shrinkage (volumic contraction) during polymerization
- supramolecular complexation via macrocyclic effect
- access to sequence control, particularly alternating copolymers



Synthesis

Mostly from ring-closing monomers, also known as cyclopolymerizable monomers

The most common underlying chemistry is radical.

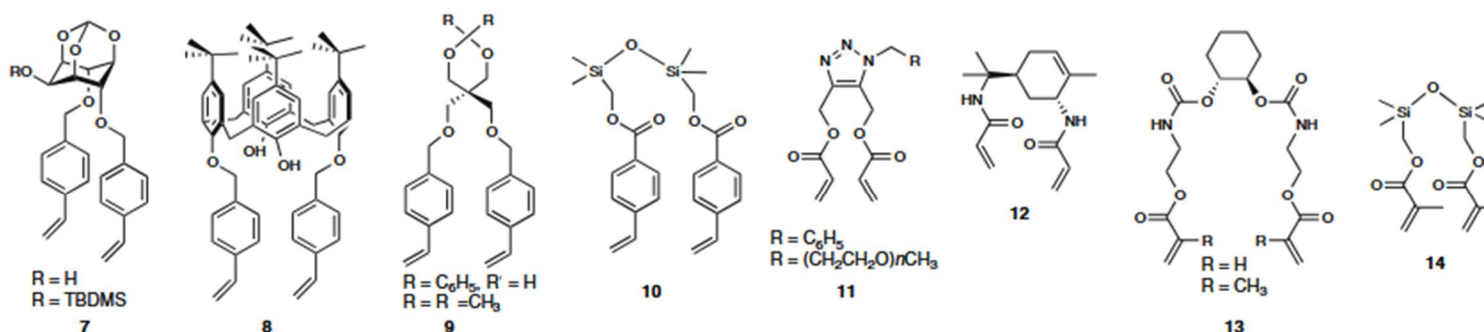


Mostly from ring-closing monomers, also known as cyclopolymerizable monomers

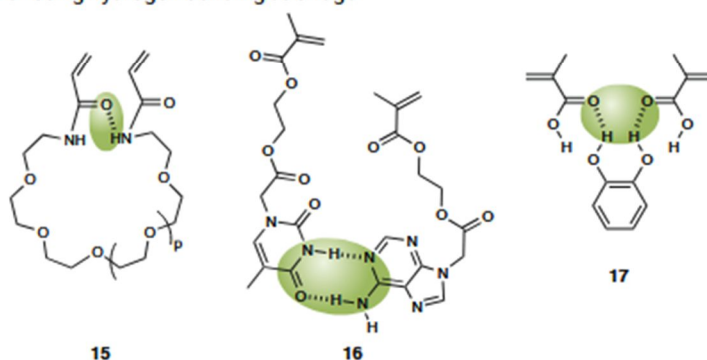
The most common underlying chemistry is radical.

B Engineering large divinyl monomers for cyclopolymerization

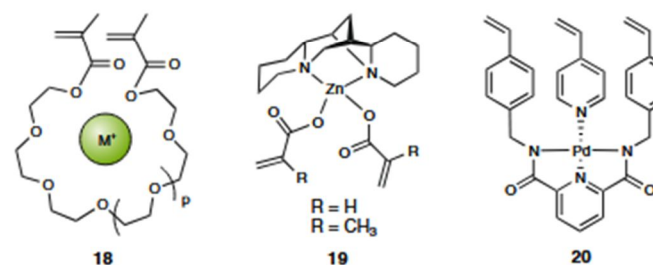
Ba Using bulky or chiral groups as template



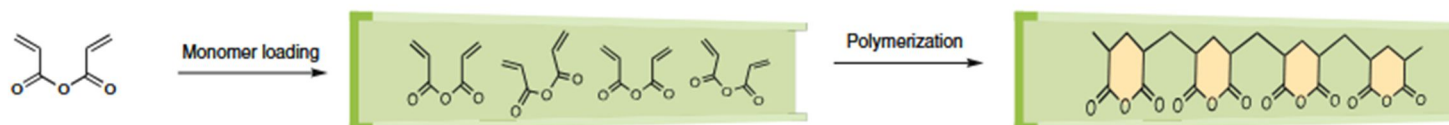
Bb Using hydrogen bonding as bridge



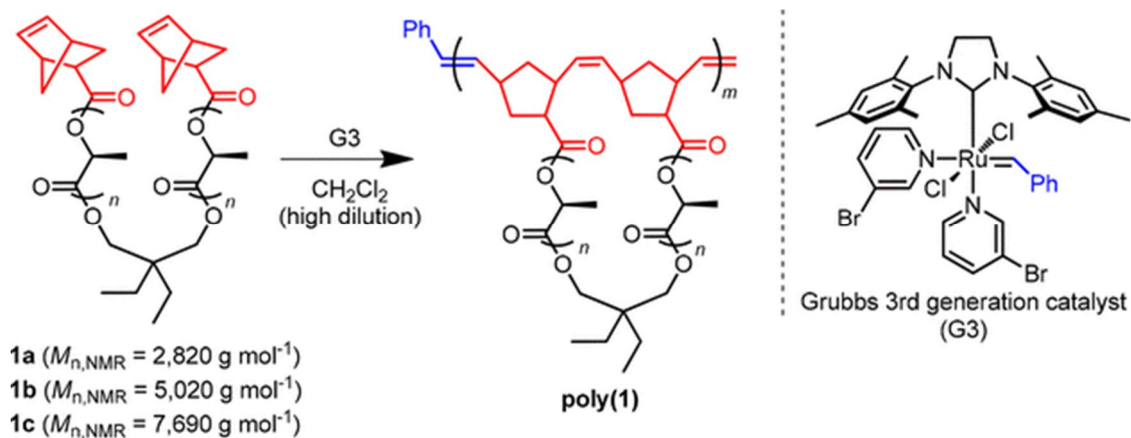
Bc Templated by metal cations



C Polymerization in nanochannels

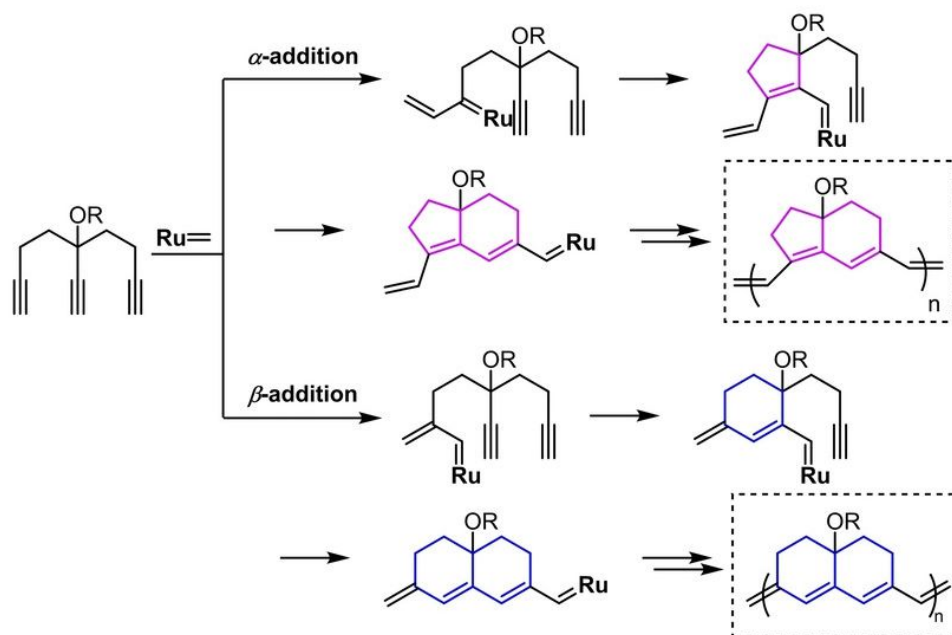


There are however further examples, not relying on radical polymerization.



ring-opening metathesis polymerization
(see Chapter 3)
ring-opening and ring-closing
simultaneously!

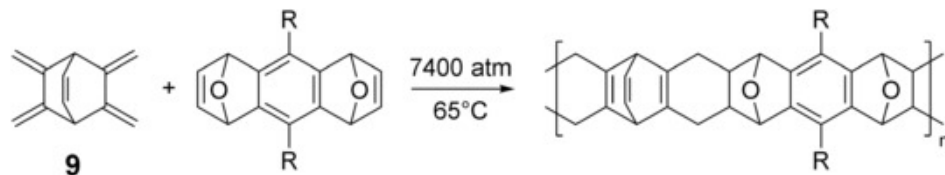
Macromolecules 2018, 51, 3855



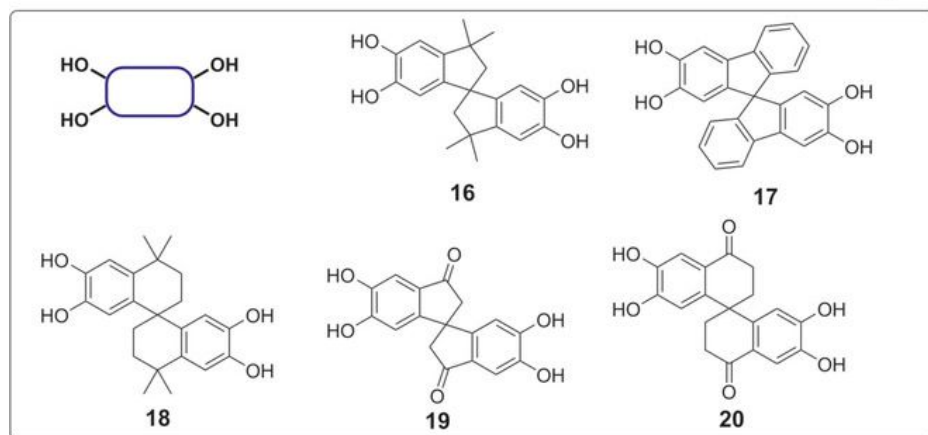
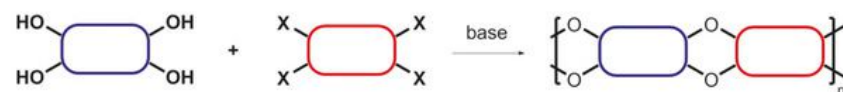
Cascade cyclopolymerization of triynes
(via metathesis)

Angew. Chem. Int. Ed. 2022, 61, e202210244

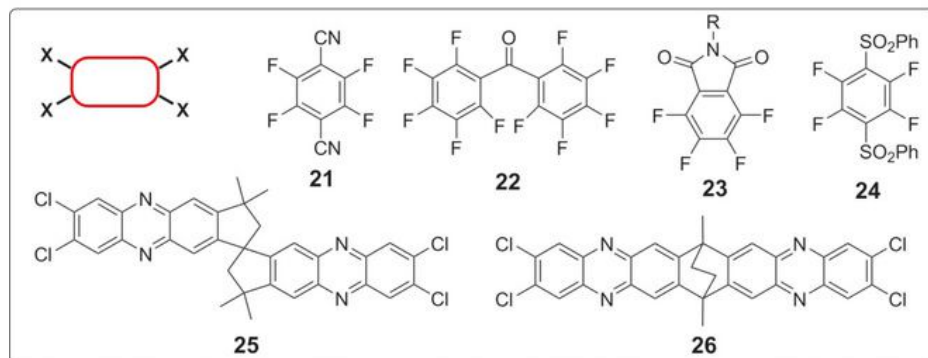
There are however further examples, not relying on radical polymerization.



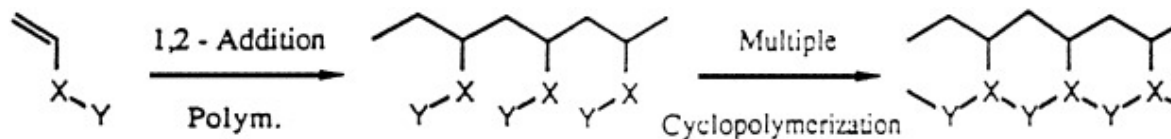
*cycloadditions
here Diels–Alder*



nucleophilic aromatic substitutions



There are however further examples, not relying on radical polymerization.



*secondary
cyclopolymerization*

